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Samara-Seed Aerodynamics

by

Eric Limacher

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Abstract

Samaras are tree seeds that spontaneously rotate about the vertical axis as they fall. This flight pattern, referred to as autorotation, generates aerodynamic lift to slow a samara's descent. In this thesis, samara flight is studied in steady-state and during a transient gust-response. In steady-state autorotation, the circulation and topology of the flow over samaras are investigated using particle image velocimetry (PIV). The stable attachment of leading-edge vortices - a lift-enhancing mechanism - is shown to depend on tip speed ratio. Leading-edge circulation, however, is insensitive to changes in tip speed ratio. The dynamic gust-response of samaras is studied using high-speed video. A linear dynamic model is developed based on momentum theory, and is shown to adequately predict the post-gust acceleration of an *Acer negundo* samara. The model predicts that axial gusts of sufficient strength or frequency can disrupt autorotation by compromising pitch-stability.

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Table of Contents

Abstract	ii
Acknowledgements	iii
Table of Contents	iv
List of Tables	vi
List of Figures	vii
List of Symbols	xi
1 Introduction	1
2 Samara flight and ecology	5
2.1 Introduction to seeds	5
2.2 Ecology of samaras	6
2.3 Evolutionary history and optimization of samara flight	8
2.4 Equilibrium flight parameters of samara flight	10
2.5 Stability of single-axis autorotation	12
2.6 Leading-edge vortices in samara flight	13
3 On the distribution of leading-edge vortex circulation in samara-seed flight	15
3.1 Introduction	15
3.2 Experimental setup	19
3.3 Results	24
3.3.1 Circulation on span-normal planes	24
3.3.2 Flow topology	24
3.3.3 Shear-layer measurements	27
3.4 Discussion	30
3.5 Conclusions	35
4 Robust gust-response of autorotating samaras	37
4.1 Introduction	37
4.2 Theoretical model	40
4.3 Methods	44
4.4 Results	46
4.5 Discussion	50
4.5.1 Application of momentum theory to the transient dynamics of autorotation	50
4.5.2 Consequences for samara ecology	54
4.6 Conclusions	55
5 Conclusions and outlook	56
5.1 Leading-edge vortices in samara flight	56
5.2 Gust-response and stability of samara flight	60
5.3 Closing remarks	62
A Sources of error and uncertainty in the PIV investigation	67
A.1 In-plane particle displacement	67
A.2 Out-of-plane particle losses	68
A.3 Particle density	69
A.4 Correlation peaks and rejected vectors	71

A.5	Estimate of uncertainty in velocity, vorticity and circulation	71
B	Estimation of the uncertainty in the reported angles of attack	78
C	MATLAB code for the calculation of leading-edge circulation	82
D	MATLAB code for extracting rotational frequency of an autorotating samara from high-speed video	87

List of Tables

2.1	Flight parameters of descending samaras from various species, extracted from several sources. The column labelled <i>Axes</i> refers to the number of autorotational axes. Samaras that autorotate about two axes exhibit notably low optimality ratios in quiescent air.	11
4.1	Equilibrium tip speed ratios of various samara species, as extracted from the literature. The observed natural range of tip speed ratio is similar to that shown in chapter 3 to allow for stable LEV attachment in steady-state. . . .	41
4.2	Physical properties and equilibrium kinematics of <i>Acer negundo</i> test samara. Kinematic parameters were extracted from the quiescent test case.	47
A.1	Frame rate and typical particle displacement (in pixels) for both FoVs. Typical pixel displacement is calculated as per equation A.1.	68
A.2	Q-factor values and number of rejected vectors for spot-checked PIV correlations.	71
A.3	Uncertainties in velocity and normalized vorticity for the large FoV test cases. The worst-case uncertainty in velocity is 0.6% of the freestream velocity, and the worst-case uncertainty in normalized vorticity is 2.4% of the peak negative normalized vorticity on any span-normal plane.	74
A.4	Measured values of normalized leading-edge circulation, Γ^* , and associated uncertainty, u_Γ^* . The worst-case relative uncertainty is 0.7% of the calculated value of Γ^*	76
B.1	Uncertainty in measured twist angle as a function of span-wise position . . .	79

List of Figures and Illustrations

2.1	Dispersal curve comparison of maple and ash samaras. The skew of the ash dispersal curve towards greater distance is explained by Green (1983) as being due to the relative scarcity of safe germination sites as compared to the maple tree. (Figure adapted from Green (1983))	7
2.2	(a) Schematic of a falling samara, labelling the geometric and kinematic parameters in equilibrium autorotation. (b) The stable pitch angle of a samara is measured relative to the swept cone. For clarity, an exaggerated positive pitch angle is shown; the range of pitch angles reported in the literature is actually slightly negative, lying between 0° and -3°	10
3.1	Mean flow topology over a rotating flat plate at a given span-wise position. For a non-zero V_∞ , the mean wake must transition from a bluff-body wake to a bound LEV topology as tip speed ratio increases, such that the full-saddle point, S, is replaced by a half-saddle, H. Note that the effective angle of attack, α , decreases with increasing tip speed ratio. A twisted plate will be compared to a flat plate in the experimental investigation in order to decouple the effects of tip speed ratio and angle of attack.	18
3.2	(a) Flow configuration for experiments. V_∞ and Ω are varied to achieve different tip speed ratios, λ . The plate-fixed coordinate system is centred at the leading edge in line with the root of rotation as shown here. (b) The dimensions of the chosen planform are shown here. Velocity fields on five span-normal planes were studied using particle image velocimetry (PIV). . .	20
3.3	(a) Test apparatus set up for dye visualizations. Rotation of the plate is driven by a chain drive connected to an electric gearmotor above the waterline. (b) PIV setup in the water-tunnel. Two photos are taken once per rotation when the span-wise axis of the plate is perpendicular to the laser sheet. Five span-normal planes at different span-wise positions were sampled for each test case.	21
3.4	Using a 105mm lens, a small FoV was used to quantify shear-layer vorticity flux on span-normal planes. The experiment was then repeated with a 60mm lens to capture a larger FoV in order to determine the overall flow topology, and to quantify the mean circulation.	22
3.5	Photo of the twisted and flat plates. The twisted plate is twisted about the leading edge (span-wise axis) as shown. Twist profile $\tau(z)$ is chosen such that the α -distribution of the twisted plate at $\lambda = 2.8$ is equal to that of the flat plate at $\lambda = 1$ (see figure 3.6).	22
3.6	(a) Twist, τ , versus normalized span-wise position, z/R , for the three test cases. Twist is measured relative to the swept disk of the plate. (b) Local effective angle of attack, α , versus z/R for all three test cases. $\tau(z)$ for the twisted plate was chosen specifically to cause the shown collapse of its α -distribution to that of the flat plate at $\lambda = 1$	23

3.7	Normalized leading-edge circulation, Γ^* , versus local effective angle of attack, α . Trailing edge separation is not accounted for in the calculation of Γ^* . For each test case, each data point represents a different span-wise location. Γ^* collapses all along the span for the twisted plate at $\lambda = 2.8$ and the flat plate at $\lambda = 1$, showing Γ^* to be independent of λ . For constant $\lambda = 2.8$, an increase in α at each span-wise location causes an increase in Γ^*	25
3.8	Contours of normalized span-wise vorticity, ω^* , from the mean velocity fields. Each column represents a different test case (different tip speed ratio and twist profile), as labelled at the top. Each row represents a different span-normal plane, moving in an outwards span-wise direction from top to bottom. A more compact recirculation zone is observed at inboard planes for higher λ , which is attributed to the increase in outwards span-wise flow. The white rectangle in each image represents the plate, and the black regions are in shadow. The plate moves from right to left through the FoV, and the freestream flow is from the bottom to the top of the FoV.	26
3.9	Chord-normal velocity is sampled along a horizontal line extending from the leading edge of the suction side of the plate. Vorticity flux through the shear layer, J_s , is the integral across the shear layer of chord-normal velocity times span-wise vorticity (see equation 3.9). Vorticity contours are shown in blue to highlight the rotational fluid in the shear layer.	28
3.10	Normalized shear-layer velocity profiles, $v_\perp/\overline{v_{\text{eff}}}$ for (a) the flat plate at $\lambda = 1$, (b) the flat plate at $\lambda = 2.8$, and (c) the twisted plate at $\lambda = 2.8$. Velocity is sampled along a line colinear with the suction side of the plate on each test plane (see figure 3.9), as denoted by the coordinate ξ . Normalized peak velocity is relatively flat along the span for each test case, and is equal to about 1.3 for all test cases. Shear-layer thickness is greatest for the twisted plate at $\lambda = 2.8$, leading to greater shear-layer vorticity flux.	29
3.11	(a) Normalized vorticity flux through the leading-edge shear layer, J_s^* , versus normalized span-wise position, z/R . Twist profile appears to govern shear-layer vorticity flux, as the data for both flat plate test cases collapse independently of tip speed ratio. (b) Normalized ratio of shear-layer flux to circulation, plotted versus z/R . No obvious collapse is observed, showing that there is no clear correlation between shear-layer vorticity flux and steady-state circulation.	31
3.12	Topological transition in the two-dimensional parameter space of tip speed ratio, λ , and normalized span-wise position, z/R . A minimum tip speed ratio is necessary to generate the bound LEV topology, which is then observed only at span-wise positions sufficiently close to the centre of rotation. In zero freestream conditions, where tip speed ratio approaches infinity, the transition is expected to approach a maximum span-wise position for a given planform. Effective angle of attack appears to have a lesser effect on the span-wise location of the topological transition than tip speed ratio.	32

4.1	(a) Mechanism of pitch-stability in samara flight based on the alignment of the weight, mg , and the aerodynamic force, F_n , as proposed by Norberg (1973). The diagram shows a cross-section of a samara on a span-normal plane. (b) An increase in pitch angle causes the centre of pressure to move towards the trailing edge, creating a restoring moment to reduce the pitch angle. (c) A decrease in pitch angle has the opposite effect, causing a restoring moment in the reverse direction. (Graphic adapted from Norberg (1973))	39
4.2	(a) Cartoon of the three-dimensional flow-field around an autorotating samara, as investigated previously by Lentink et al. (2009) and Lee et al. (2014). The presence of a stably-attached LEV enhances force production, and is hypothesized herein to stabilize the instantaneous CoP location to improve pitch-stability. (b) The flow-field as observed on a span-normal plane at a sufficiently inboard location. The LEV core does not convect into the wake, but remains in place relative to the samara.	41
4.3	Schematic of an autorotating samara. To model an axial gust, the freestream velocity, V_∞ , will be treated as an input to the system. The time constant of acceleration associated with the gust-response will be derived using momentum theory.	42
4.4	(a) Diagram of the experimental setup. The samara was dropped from sufficient height so as to allow the samara to reach equilibrium kinematics prior to hitting the transverse gust. A video of the samara in quiescent air was used to establish the equilibrium kinematics. (b) <i>Acer negundo</i> samara used in the videos. The span-wise CoR location was determined from the video taken in quiescent air. Measured values of total span, b , maximum chord length, $c_{\max}$, and span-wise distance from the CoR to the tip, R , are shown.	45
4.5	Superimposed images from the videos of the samara (a) falling in quiescent air and (b) falling through a transverse gust from left to right. The equilibrium kinematics of the samara are extracted from the quiescent test case. The inertial response of the samara to the transverse gust is captured by tracking the post-gust displacement of the samara's estimated CoM. The beginning of the post-gust response is taken as the point in time where the CoM is observed to stop momentarily, labelled in the figure as point A.	46
4.6	(a) The post-gust response of the samara is shown with superimposed frames from the video separated in time by 2.5ms. The CoM location was manually tracked every 5ms to produce the original estimates. A parabolic arc was fitted to these estimates, and a set of refined CoM location estimates were taken to lie on this arc. (b) The post-gust displacement, as extracted from the figure in (a), plotted versus time. Unsteady momentum theory overpredicts the observed inertial time constant by 20%, as determined by exponential regression of the experimental displacement data. The uncertainty in the displacement, as shown with errorbars, is based on the variance of the original estimates relative to the parabolic curve fit in (a).	49

4.7	The deviation of tip speed ratio from equilibrium, λ/λ_0 , in response to the shown axial gust profile, V_∞/v_0 . When the inertial time constant is much less than the gust period, i.e. $\tau_i/\tau_g \ll 1$, tip speed ratio does not deviate greatly from equilibrium. When the gust period is of similar magnitude or less than the inertial time constant, significant deviations from equilibrium occur, which is liable to disrupt autorotation by compromising pitch-stability.	53
5.1	Comparison of vorticity distributions and mean flow topology for different tip speed ratios at the most inboard test plane. Increasing tip speed ratio stretches the leading-edge vorticity distribution, making it more compact without changing the integral value of circulation. A minimum tip speed ratio is necessary to generate the bound-LEV topology. The local effective angle of attack, α , is the same for the two test cases shown. Blue and red contours represent opposite-signed vorticity.	57
5.2	Cartoon of the flow-field over a rotating plate exhibiting a stable LEV. Tip speed ratio affects the span-wise extent of the bound-LEV topology; further work is necessary to adequately predict the span-wise extent of the bound-LEV topology under general conditions.	58
5.3	At a given span-wise position, local effective angle of attack and local effective velocity determine the global circulation in attached flow (see right); the same inflow parameters determine leading-edge circulation in the separated flows of the current investigation, as calculated from the mean velocity field (see left). Tip speed ratio is important in determining the location of the saddle point.	59
A.1	Sample images from the twisted plate test case at $\lambda = 2.8$ at the midspan location. Left column: Small FoV used to quantify shear-layer velocity profile. Right Column: Large FoV used to quantify leading-edge circulation. Top Row: 800x800 pixel window. Red rectangle highlights the region that is portrayed in the middle row. Middle Row: 128x128 interrogation window, as sampled in the first pass of the multi-pass correlation scheme. The red rectangle in these images highlights the region shown in the bottom row. Bottom Row: 16x16 interrogation window; this is the smallest interrogation window used. At least four local maxima of brightness are visible for both FoVs, demonstrating sufficient particle density to acquire accurate correlations.	70
A.2	Normalized circulation, Γ^* , versus local effective angle of attack, α . The vertical error bars denote the estimated uncertainty in the normalized circulation, as calculated using equations A.10 and A.15. The error bars are smaller than the data markers, which makes them difficult to distinguish.	76
B.1	Normalized circulation, Γ^* , versus local effective angle of attack, α . Uncertainty in the reported angle of attack is represented by the horizontal error bars.	81

List of Symbols, Abbreviations and Nomenclature

Abbreviation

AR	aspect ratio
CoM	centre of mass
CoP	centre of pressure
CoR	centre of rotation
FoV	field of view
LEV	leading-edge vortex
MAV	micro-aerial vehicle

Definition

Symbol

b	total span
c	chord length
C	image calibration constant in metres per pixel
$\bar{c}$	average chord length = wing area / span
$c_{\max}$	maximum chord length
J_s	shear-layer vorticity flux
K	steady-state post-gust velocity magnitude
m	samara mass
m_a	apparent mass associated with an accelerating samara
R	span-wise distance from CoR to tip
R_d	radius of the swept disk
Re	Reynolds number
s	post-gust displacement of a samara
S	swept disk area
t	time
T	thrust acting normal to an actuator disk
v	post-gust velocity of a samara
v_d	descent velocity
v_0	equilibrium descent velocity of a samara
v_i	induced velocity
v_{eff}	local effective velocity
$\overline{v_{\text{eff}}}$	average effective velocity along the span
v_g	gust velocity magnitude
v_{mid}	effective velocity at midspan
v_r	fluid velocity relative to swept disk
v_s	mean sinking velocity of samara during descent
v_w	mean horizontal wind speed
$v_{\perp}$	plate-normal velocity
V_{∞}	freestream velocity
x, y, z	coordinates in plate- or samara-fixed frame of reference
x_h	horizontal dispersal distance of a samara

w	span-wise velocity
W	samara weight
α	local effective angle of attack
β	coning angle
δ	uncertainty in samara CoM location
Γ	circulation
λ	tip speed ratio
λ_0	equilibrium tip speed ratio
ν	kinematic viscosity
ρ	fluid density
ξ	shear-layer coordinate ahead of leading edge
τ	plate twist profile
τ_g	gust period
τ_i	inertial time constant of accelerating samara
Φ	vorticity annihilation
ω	vorticity
Ω	rotational speed
*	superscript denoting normalization by local parameters
**	superscript denoting normalization by global parameters

Chapter 1

Introduction

Nature is a boundless source of inspiration. To the mechanical engineer, nature is full of complex machines that elegantly achieve their functions. *Biomimicry* involves copying mechanisms, structures or processes found in nature into the design of engineered devices. The organisms that inhabit any ecosystem have adapted over countless generations to navigate the shifting opportunities and dangers of their environment. Such adaptations lead to varied degrees of reproductive success, which is the key determinant of the behavioural and morphological features that persist within a species. Although unknown to the organisms themselves, their inherited forms and instinctual behaviours take advantage of underlying physical mechanisms. Understanding these mechanisms can inspire engineers to design novel systems in a variety of fields.

This thesis looks to nature in order to better understand the aerodynamics of small-scale flight. The investigation focuses on *samaras*, which are tree seeds that spontaneously settle into a rotating pattern as they fall from their parent organisms. This kinematic pattern exhibited by samaras is referred to as *autorotation* throughout this document, although it may be considered *pseudo-autorotation* by some classical definitions (Lugt, 1983). Samaras of different species can rotate about one or more axes, but the current work focuses primarily on those that rotate about the vertical axis only.

A review of samaras in their natural context led to the aerodynamic research questions addressed in this thesis. Chapter 2 explores the role and function of samaras in the natural realm through a review of the ecological literature. The evolutionary history of samaras is briefly discussed, followed by a discussion of how natural selection has optimized samara flight since its emergence. In order to facilitate further aerodynamic investigation, samara

flight is abstracted to a set of dimensionless parameters. Tip speed ratio was selected for further study, and the range of tip speed ratios exhibited by a variety of samara species was extracted from the literature.

Of particular importance to small-scale flight is the formation of stably-attached leading-edge vortices (LEVs). Within the context of this work, the terms *stable* or *stably-attached* refer to LEVs which maintain a convection speed of zero relative to the rotating wing. LEVs have been identified as a lift-enhancing mechanism in insect flight (Ellington et al., 1996; van den Berg and Ellington, 1997), and have since been observed in the flight of several samara species (Lentink et al., 2009; Lee et al., 2014). Much research has already been conducted on the stable attachment of LEVs over impulsively-started rotating plates in quiescent fluid, e.g. Jones and Babinsky (2011); Garmann et al. (2013); Venkata and Jones (2013); Wojcik and Buchholz (2014). In the flight of samaras, however, their descent introduces a freestream velocity parallel to the axis of rotation which has not been studied previously. Chapter 3 presents an experimental investigation of rotating plates in a steady freestream flow, serving as an abstraction of natural samara flight. The independent effects of tip speed ratio and angle of attack on the flow-field are systematically studied using particle image velocimetry (PIV) on span-normal planes. Forces are not directly measured, so flow topology and circulation are studied as proxies for force. The leading-edge shear layer is also investigated in an attempt to explain the observed trends of steady-state circulation. One long-term goal of such studies is the development of a comprehensive force model for propellers in deep stall, accounting for the formation of coherent vortical structures.

Having established the ecological merits of autorotation, and having gained some understanding of the steady-state flow-field inherent in samara flight, attention is turned to the transient dynamics of autorotation. In chapter 4, the gust-response of an *Acer negundo* samara is investigated using high-speed video. In response to a horizontal gust, the test samara is shown to continue its autorotation as it undergoes a sharp change in trajectory.

A dynamic model is developed based on momentum theory and is validated against the observed gust-response of the test samara. The implications of the model on the limits of stable autorotation are then discussed, and the effects on stable LEV attachment are also considered. The developed linear model is not specific to samaras; it could be applied to other small (low-inertia) propellers in unsteady flows.

Further introduction of the experimental investigations of chapters 3 and 4 is given at the end of chapter 2, after the review of samaras in nature has provided a unifying context. Chapters 3 and 4 are both written in the format of journal articles, as the present thesis is manuscript-based. Chapter 3 was submitted for review to the *Journal of Fluid Mechanics* on October 10, 2014 under the title *On the distribution of leading-edge vortex circulation in samara-seed flight* (Submission number JFM-14-S-1070). It has been copied into this thesis, with changes made as necessary to abide by the University of Calgary's thesis formatting guidelines, and to maintain consistent nomenclature throughout this document. Chapter 4 has been prepared for submission to the journal *Bioinspiration and Biomimetics*, but as it references the former work, its submission will wait until such time that the former work is accepted for publication. The author of the current thesis is the first author on both of these journal manuscripts; Dr. David E. Rival, supervisor for this MSc. thesis, is second author on both manuscripts.

Some supporting content could not be included in the concise manuscript format of chapters 3 and 4, but it is included in the appendices for completeness of this thesis. Appendix A discusses the sources of error and measurement uncertainty in the PIV investigation of chapter 3; appendix B presents the uncertainty analysis for the reported angles of attack in the same investigation. The MATLAB code used to calculate leading-edge circulation from the PIV correlations is included in Appendix C. Appendix D provides MATLAB code to automatically extract the rotational frequency of an autorotating samara from high-speed video.

Chapter 5 summarizes the findings of the full work, including those obtained through literature review and experimental investigation. Connections between the stand-alone works of chapters 3 and 4 are discussed and related to biomimetic applications where appropriate. As the limits of the current scope are identified, recommendations for future work are given.

Chapter 2

Samara flight and ecology

2.1 Introduction to seeds

The dispersal of seeds is a ubiquitous reproductive strategy of plants that is achieved in a variety of ways. Many plants will produce a large quantity of seeds in order to maximize the probability that one will end up in a suitable site to germinate; other plants will produce fewer seeds, but will invest more mass in each seed to maximize its individual probability of survival. Henery and Westoby (2001) performed a study on 47 species of plants across 14 families, and found an inverse relationship between the number of seeds produced by a tree and the average mass of each seed. The total mass of all seeds produced represents an expenditure of energy, and each tree can only dedicate a portion of its energy to reproductive purposes. It can be said that this trade-off has been well explored through the evolutionary divergence within the plant kingdom, with reproductive propagules ranging from the miniscule seeds of orchids (as little mass as 0.0001 mg) to the 20-kg fruit of the double coconut (Moles et al., 2005).

Plants can disperse their fruit by a variety of mechanisms, relying on living (biotic) or non-living (abiotic) elements of their environment. Fruit that is dispersed by the wind tends to have lower mass than fruit that relies on biotic means or gravity alone; one could hardly imagine coconut trees relying on the wind to disperse their fruit. Among plants that rely on wind dispersal, there is still significant spread in seed mass. The specimens studied by Minami and Azuma (2003) ranged in mass from 0.02mg (*Solidago altissima*) to 14.3mg (*Tabebuia chrysotricha*). These two species employ different aerodynamic mechanisms to achieve dispersal from the parent organism. The seeds of *Solidago altissima* are pappose-style seeds, comprising a plume of slender hairs that sum to a large surface area, thereby

generating high viscous drag. The seeds of *Tabebuia chrysotricha* are winged, generating lift as they fall in order to slow their descent. By artificially altering the weights of pappose and winged diaspores from different species, Matlack (1987) found a critical seed mass above which the winged seeds under study achieved slower terminal velocities than pappose seeds. Slower seed descent velocity is a boon to the reproductive success of wind-dispersed seeds, which explains the prevalence of winged morphologies for larger seed masses. The benefit of reduced descent velocity will be discussed in further detail in the following section.

2.2 Ecology of samaras

Samaras represent a subset of winged seeds that rotate about the vertical axis as they descend. This kinematic pattern produces a high amount of lift, reducing the sinking velocity of the samara. Slower descent translates into greater potential dispersal distance, as the seed is exposed to horizontal winds for a longer period of time. This phenomenon is described by the following relationship from Greene and Johnson (1990):

$$x_h = h \frac{v_w}{v_s}, \quad (2.1)$$

where x_h is the horizontal distance travelled by the seed, h is the height from which it abscises from the parent tree, v_s is its mean sinking velocity during its descent, and v_w is the mean horizontal wind velocity. Generally speaking, greater dispersal distance leads to greater probability that one or more seeds will find an appropriate site to germinate and survive into adulthood.

It is also beneficial for a source tree to achieve a spread of seed-dispersal distances to colonize as many suitable sites as possible. Greene and Johnson (1992) concluded that variation of seed mass or wing loading is unlikely to be the key driver in varying dispersal distance, stating that the variations due to wind turbulence are likely to be far greater than that which could be achieved through morphological variations. The work of Nathan et al.

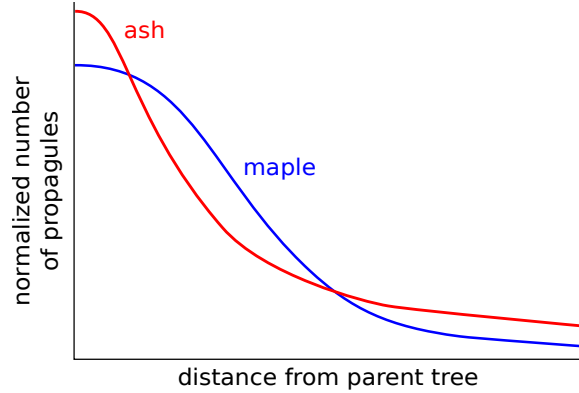


Figure 2.1: Dispersal curve comparison of maple and ash samaras. The skew of the ash dispersal curve towards greater distance is explained by Green (1983) as being due to the relative scarcity of safe germination sites as compared to the maple tree. (Figure adapted from Green (1983))

(1996) with *Pinus halepensis* affirmed this theory; while significant variations in morphology were observed, a much smaller variation in sinking velocity was observed.

Whereas one may not find major variation in sinking speed amongst the seeds of a particular species, comparisons across species yield significant variation. Natural selection can act on the mean sinking velocity of a species' propagules to alter the shape of the dispersal curve. Green (1983) postulated that if the density of safe sites around the parent plant is sparse - that is, sites suitable for seed germination are rare - then there is a selective pressure to achieve greater dispersal distances. Conversely, if safe sites are abundant, a dispersal distribution that is skewed to shorter distances can result in greater numbers of offspring successfully germinating, surviving and subsequently reproducing. Green (1983) compared the dispersal curves of maples (*Acer spp.*) and ash trees (*Fraxinus spp.*). These trees co-exist in the forests of eastern North America, but their saplings have different tolerances for environmental conditions. In reference to earlier work, Green (1983) notes that maple saplings can thrive under smaller gaps in the tree canopy overhead than those of *Fraxinus spp.*; within the same forest, maple trees therefore experience a greater density of safe sites for their offspring than ash trees. Green (1983) goes on to argue that this explains the greater density of *Fraxinus spp.* propagules at greater distances, as shown in figure 2.1.

2.3 Evolutionary history and optimization of samara flight

The process of natural selection optimizes certain characteristics of organisms that enable them to reproduce more consistently and prolifically. In the case of samara-producing trees, whose reproduction depends on seed flight, it stands to reason that evolution would optimize the seed flight in some way. If the mechanism of autorotational flight has evolved independently in unrelated organisms, it stands out as a solution to a common problem. Matlack (1987) asserted that samaras are indeed an example of convergent evolution. He notes that in different genera, the winged portion of the samara develops differently, indicating different evolutionary histories. In *Acer* species, the winged portion of the samara is formed as an extension to the ovary wall prior to fertilization; in *Liriodendron* species, the samaras are two fused carpels, and in *Carpinus* species, the winged portion is a leafy bract (specialized leaf). Despite the developmental differences among these genera, all of their fruit exhibit autorotational behaviour. Samaras are found among a wide range of plant families, which tends to align with the hypothesis of convergent evolution. Samaras are even found amongst both gymnosperms and angiosperms, where angiosperms include the vast variety of all flowering plants seen today.

What reasons can be postulated for the autorotational flight mechanism to have evolved multiple times? In reference to the trade-off described earlier, the autorotational mechanism is well suited to distributing more mass to each individual seed, while still achieving sufficient dispersal distance from the parent tree. Drag-based pappose seeds can achieve long dispersal, but can invest very little mass in each seed. Much heavier fruit can rely on biotic dispersal by animals, but this strategy requires its own set of morphological adaptations with associated energy costs. Samara flight occupies a niche between these two extremes, allowing an intermediate quantity and mass of seeds to be produced. Intense interspecific competition in marginal soil conditions could favour this balanced approach. When safe sites for germination are rare, a species can increase its chances of colonizing those sites by

increasing the number of seeds produced. On the other hand, a sapling in a competitive environment must outcompete its rivals, requiring sufficient nutrients upon germination to establish itself sooner.

Through natural selection, further optimization of this kinematic pattern is expected to have occurred since its emergence. Optimal descent velocities for samaras were considered by Azuma and Yasuda (1989). They studied several species of autorotating seeds, recording their steady-state descent velocities in quiescent air. In order to reach this terminal velocity, the weight of the falling seed must be balanced by the rate of change of momentum of the surrounding air. The energy required to accelerate the fluid comes from the gravitational potential energy of the samara, such that a non-zero minimum sinking speed must exist. By treating a samara as an actuator disk and applying momentum theory, Azuma and Yasuda (1989) derived the following expression for optimal velocity, v_{opt} :

$$v_{opt} = \sqrt{\frac{2}{\rho}(W/S)}, \quad (2.2)$$

where ρ is the density of air, W is the weight of the seed, and S is the area of the swept disk (the disk scribed by the tip path). Azuma and Yasuda (1989) documented a spread of sub-optimal descent velocities in the seeds they studied, through which a trend towards optimality can be discerned; this trend is driven by the energy constraints of the parent organism. An optimal autorotor achieves the smallest swept area for a given descent velocity and seed mass. The swept area is correlated to the size of the winged portion of the samara, which does not translate into better survival once the seed has landed on the ground. Ideally, as little mass as possible is invested in the wing in order to provide maximum nutrient mass to the new sapling instead. However, a minimum wing area is necessary to maintain the appropriate mass balance for stable autorotation, which may limit the optimality exhibited by samaras.

2.4 Equilibrium flight parameters of samara flight

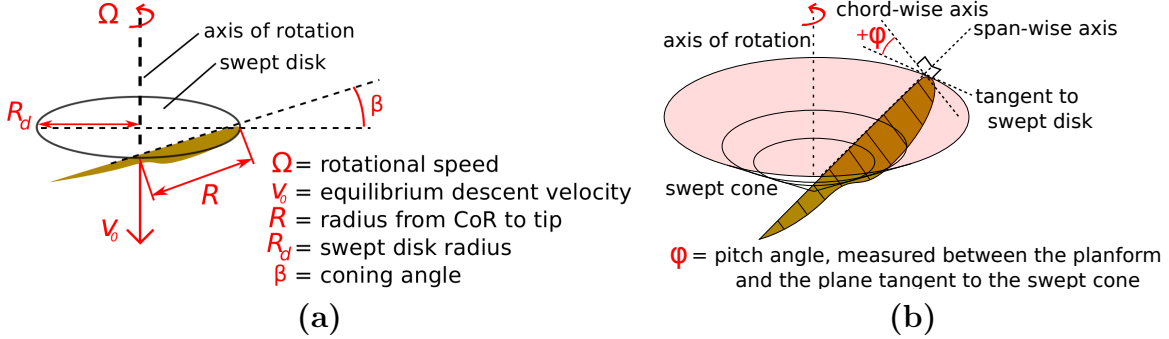


Figure 2.2: (a) Schematic of a falling samara, labelling the geometric and kinematic parameters in equilibrium autorotation. (b) The stable pitch angle of a samara is measured relative to the swept cone. For clarity, an exaggerated positive pitch angle is shown; the range of pitch angles reported in the literature is actually slightly negative, lying between 0° and -3° .

In order to study samaras further for biomimetic purposes, samara flight will be abstracted to a tractable aerodynamic problem. A samara dropped from a sufficient height will enter autorotation and establish equilibrium kinematics, reaching constant values of descent velocity, v_0 , rotational speed, Ω , and coning angle, β (measured relative to the swept disk), as depicted in figure 2.2a. The equilibrium pitch angle is measured relative to the plane tangent to the swept cone, as depicted in figure 2.2b. Equilibrium pitch angles tend to lie between 0° and -3° (Azuma and Yasuda, 1989; Lentink et al., 2009). Azuma and Yasuda (1989) showed coning angle to vary more, ranging from 13° to 35° in the ten species they studied. The aerodynamic effects of coning angle will not be investigated in the current work, although it could represent a rich topic for further study. If we consider only cases where coning angle and pitch angle are held at zero, the kinematics of a single-axis autorotor in quiescent air can be characterized by two dimensionless parameters: tip speed ratio, λ , and Reynolds number Re , defined as

Species	λ	Re	v_0	v_{opt}/v_0	Axes	Source
	—	—	(m/s)	—	—	
<i>Acer platanoides</i>	3.36	2482	0.90	0.83	1	Norberg 1973
<i>Picea abies</i>	2.16	326	0.64	0.92	1	Norberg 1973
<i>Acer diabolicum</i> Blume	3.54	1690	0.82	0.74	1	Azuma 1989
<i>Acer palmatum</i> Thunb.	2.08	635	1.09	0.64	1	Azuma 1989
<i>Acer palmatum</i> Thunb. Matsumurae Makino	2.32	1160	1.04	0.65	1	Azuma 1989
<i>Carpinus Tschonoskii</i> Maxim	1.74	832	1.02	0.61	1	Azuma 1989
<i>Pinus Thunbergii</i> Parlatore	2.82	972	0.98	0.62	1	Azuma 1989
<i>Cedrus deodora</i>	2.99	2441	0.84	1.12	1	Minami 2003
<i>Pinus luchuensis</i>	2.59	516	0.78	0.60	1	Minami 2003
<i>Fraxinus japonica</i> Blume	1.44	1016	1.72	0.38	2	Azuma 1989
<i>Liriodendron tulipifera</i> L.	1.51	1079	1.19	0.39	2	Azuma 1989

Table 2.1: Flight parameters of descending samaras from various species, extracted from several sources. The column labelled *Axes* refers to the number of autorotational axes. Samaras that autorotate about two axes exhibit notably low optimality ratios in quiescent air.

$$\lambda = \frac{\Omega R_d}{v_0}, \text{ and} \quad (2.3)$$

$$Re = \frac{\bar{c} \sqrt{v_0^2 + (\Omega R_d)^2}}{\nu}, \quad (2.4)$$

where R_d is the radius of the swept disk traced by the tip, ν is the kinematic viscosity of air, and $\bar{c}$ is the average chord length, defined as wing area over span. For several species of samaras studied by Norberg (1973), Azuma and Yasuda (1989) and Minami and Azuma (2003), these dimensionless flight parameters are compared alongside the measured descent velocities and the optimality ratio, v_{opt}/v_0 . Optimal descent velocities are calculated from the published values of disk-loading using equation 2.2, assuming the density of air to be $\rho = 1.2\text{kg/m}^3$.

The two species with the highest autorotative optimalities in table 2.1 are both conifers: *Cedrus deodora* and *Picea abies*. The former actually displayed an autorotative optimality greater than unity, which either points to measurement error, or to the limitations of the simplifying assumptions in the derivation of equation 2.2. Even if the underlying assumptions

do not hold generally true, the optimal descent velocity remains a useful comparative metric.

2.5 Stability of single-axis autorotation

The optimalities of the two-axis samaras (*Fraxinus japonica* and *Liriodendron tulipifera*) are conspicuously low, and their descent velocities are higher than any of the single-axis samaras. This is in conflict with the previous discussion in section 2.2 regarding the greater maximum dispersal distance of *Fraxinus spp.* fruit compared to that of *Acer spp.*, since equation 2.1 predicts lower dispersal distances with higher descent velocities. All of the data in table 2.1 comes from tests in quiescent conditions, but samaras in the natural environment operate in variable wind conditions with gusts and turbulence. McCutchen (1977) hypothesized that the two-axis autorotative kinematics of *Fraxinus spp.* may provide more stable autorotation in gusty conditions. The unexpectedly far maximum dispersal of *Fraxinus spp.* samaras in natural conditions tends to affirm this hypothesis. Greene and Johnson (1990) directly tested this “McCutchen-Green” hypothesis. They compared the dispersal of single- and two-axis samaras in the same wind conditions, but found no deviation from the standard model in equation 2.1, negating the hypothesis. During their tests, they observed no disruption of autorotation for any of the single-axis specimens. In yet stronger and gustier winds, the McCutchen-Green hypothesis might hold, but it must be questioned whether such severe wind conditions are typical of the natural ecosystems inhabited by the species in question.

For single-axis autorotors, the necessity to maintain pitch-stability is a limiting factor to the stability of the autorotative flight pattern. In order to maintain a stable pitch angle, an alignment of the chord-wise centre of mass (CoM) and centre of pressure (CoP) is necessary. Norberg (1973) asserted that the CoM must lie between 27 and 35% of the chord from the leading edge if pitch-stability is to be achieved. The chord-wise CoM lies at the mid-chord in *Fraxinus spp.* samaras, such that the pitch angle is unstable and a secondary autorotative axis is established. The limits of stable, single-axis autorotation will be explored

in greater detail in chapter 4, where the effects of gusts on samara flight will be investigated. Momentum theory will be applied to the problem, as done previously by Azuma and Yasuda (1989). The resulting linear model describes the dynamic behaviour of an actuator disk in a changing freestream flow. Using high-speed video, the model will be validated by tracking the post-gust response of an *Acer negundo* samara.

2.6 Leading-edge vortices in samara flight

Norberg (1973) claimed that the flow around a samara must be separated in order for the alignment of the chord-wise CoM and CoP to be maintained through small disturbances. More recently, it has been found that the separated flow-field around samaras is characterized by a coherent vortical structure over the suction side of the wing. Lentink et al. (2009) observed stably positioned leading edge vortices (LEVs) over autorotating samaras of the *Acer* and *Carpinus* genera. The presence of an LEV elevates the lift experienced by the samara as it descends, as has also been found in insect flight (Ellington (1999), van den Berg and Ellington (1997), Usherwood and Ellington (2002)). Increased force production would allow for greater mass to be invested in each seed without an increase in descent velocity; this would certainly be beneficial to the reproductive success of samara-producing trees, as discussed previously. If natural selection has favoured stable LEV formation in this way, then the study of samara flight could prove useful in uncovering the mechanisms of stable LEV attachment.

In table 2.1, single-axis samaras were shown to operate within a relatively narrow band of tip speed ratios from 1.7 to 3.6. Given that rotation has been identified as crucial to LEV-stability in insect flight (Lentink and Dickinson, 2009), it seems likely that tip speed ratio, a dimensionless measure of rotation, would be important to stable LEV formation in samara flight. Let us consider the two extremes, where tip speed approaches zero or infinity. As tip speed ratio approaches zero, rotation ceases and no stable LEV is expected;

rather, the vortex-shedding associated with a bluff body is expected. As tip speed ratio approaches infinity, the samara becomes a rotating plate in quiescent fluid; this problem has been studied quite thoroughly, and stable LEV formation has been well documented in this configuration, e.g. Ellington et al. (1996); Usherwood and Ellington (2002); Birch et al. (2004); Harbig et al. (2013); Garmann et al. (2013). The systematic investigation of non-zero freestream velocities, as introduced by the non-zero descent velocity of samaras, is novel to the study of LEVs; as such, it forms the subject of experimental investigation in chapter 3. The parameter space for investigation will be informed by the natural range of tip speed ratio in table 2.1 at similar Reynolds numbers. Particle image velocimetry (PIV) will be conducted to elucidate the topology of the flow; the transition from vortex shedding to stable LEV attachment with increasing tip speed ratio will be investigated. Circulation will be quantified on span-normal planes, due to its importance to force production. The results of chapter 3 provide insight into the *steady-state* nature of LEVs in samara flight, which will allow for the *quasi-steady* consideration of LEV attachment in gusting conditions in chapter 4.

Chapter 3

On the distribution of leading-edge vortex circulation in samara-seed flight

Abstract

As an abstraction of natural samara flight, steadily rotating plates in a freestream flow have been studied. Particle image velocimetry (PIV) on span-normal planes has been conducted to show that increasing rotation, as captured by the dimensionless parameter of tip speed ratio, causes a transition of the mean wake topology from that of a bluff-body to that of a stable leading-edge vortex (LEV). This topological distinction allows for a discussion of force production, while avoiding the complexities of vortex identification. Despite its notable effect on topology, a change in tip speed ratio has negligible effect on leading-edge circulation at a given span-wise position, local effective angle of attack, and local effective velocity. Effective angle-of-attack distribution was held constant at different tip speed ratios by comparing rotating plates with different twist profiles. The shear-layer velocity profile at the leading edge was also resolved, allowing quantification of the vorticity flux passing through the leading-edge shear layer. Interestingly, the observed equilibrium values of circulation are not sensitive to changes in shear-layer vorticity flux.

3.1 Introduction

In some cases, it is possible for a coherent vortex to persist next to a moving body rather than convect into the wake. In insect flight, a leading-edge vortex (LEV) is formed and remains attached to the wing throughout one half wing stroke (van den Berg and Ellington, 1997). Experiments and simulations on insect wing models have shown that LEV attachment can be

extended beyond the 180° limit of a wing stroke, with some results showing stable LEVs up to 270° of stroke angle and beyond (Usherwood and Ellington, 2002; Lentink and Dickinson, 2009; Harbig et al., 2013). LEV attachment is even seen in nature to extend indefinitely, as with the stable LEV that forms over autorotating samara seeds (Lentink et al., 2009; Lee et al., 2014). The presence of an LEV over the suction side of a samara wing enhances force production on the wing (Azuma and Yasuda, 1989; Lentink et al., 2009) beyond what would be predicted from two-dimensional inclined flat plate experiments (Fage and Johansen, 1927). Force prediction for steadily rotating plates will require a model that accounts for LEVs, which in turn necessitates the prediction of when and where LEVs will persist in the steady-state sense. To this end, the circulation and flow topology over constantly rotating plates, long after the onset of rotation, will be investigated in the current study.

All of the above natural flight examples share common features: rotation as part of the wing kinematics, and low wing aspect ratios. These are conditions that have been shown to favour stable LEV attachment (Lentink and Dickinson, 2009). However, discussion in the literature often implies a dichotomy, where a stable LEV either exists or does not. The appearance of a dichotomy may stem from the fact that much of the research on rotating flat plates has been done in quiescent fluid (Jones and Babinsky, 2011; Harbig et al., 2013; Gar-mann et al., 2013). By contrast, the stable LEVs over natural samaras exist in a freestream flow parallel to the axis of rotation, due for instance to its descent. The introduction of this freestream, V_∞ , provides a continuum along which to study steady-state LEVs. Tip speed ratio, λ , emerges as an important dimensionless measure of rotation, defined as

$$\lambda = \frac{\Omega R}{V_\infty}, \quad (3.1)$$

where R is the span of the plate, and Ω is the rotational speed.

Consider first a stationary plate in a non-zero freestream, such that the tip speed ratio is zero. Rotational fluid passes through the leading- and trailing-edge shear layers and rolls up

into vortices that convect into the wake. A bluff-body wake of this type is characterized in the mean sense by a full-saddle point downstream of the plate (see figure 3.1). As tip speed ratio is increased from zero, centrifugal force acts on the stagnant fluid in the recirculation zone, driving it in an outwards span-wise direction. We expect this draining of the recirculation zone to make it more compact, drawing the saddle point towards the plate. At what point will such a flow-field exhibit a recognizable LEV, as observed in natural samara flight? For a sufficiently strong LEV, it is expected that the flow that separates from the leading edge will reattach behind the LEV. At some critical tip speed ratio, a topological transition must occur, whereby the full-saddle point in the wake is replaced by a half-saddle on the plate, and a near-Kutta condition at the trailing edge becomes possible, as shown in the far right image of figure 3.1. This topology will hereafter be referred to as a *bound LEV*. The gradual nature of this transition will be investigated using dye visualizations at four different tip speed ratios. As the flow-field transitions to that of a bound LEV, local instantaneous variations are expected to decrease. As rotation increases and span-wise convection of vorticity grows, in-plane convection of vorticity via vortex shedding can decrease, and a stable LEV is possible. In the current work, only mean flow-fields will be studied, noting that the conditions that are expected to generate the bound-LEV topology in the mean sense are the same as those that favour a stably-attached LEV (i.e. rotation, low aspect ratio).

Tip speed ratio is a coupled metric. It serves as both a dimensionless measure of rotation, and affects the span-wise distribution of local effective angle of attack, α , according to the relation

$$\alpha = \arctan\left(\frac{V_\infty}{\Omega z}\right) + \tau(z) = \arctan\left(\frac{1}{\lambda} \frac{R}{z}\right) + \tau(z), \quad (3.2)$$

where z is span-wise position, defined such that $z = 0$ is at the centre of rotation, and $\tau(z)$ represents the twist profile of the plate. In order to decouple the effects of tip speed ratio and effective angle of attack, a flat plate ($\tau(z) \equiv 0$) and a plate twisted along the span-wise axis

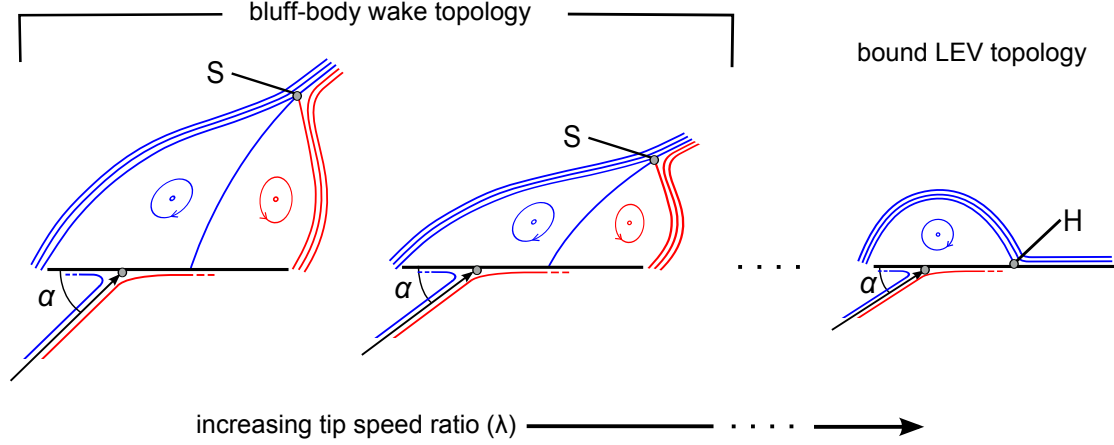


Figure 3.1: Mean flow topology over a rotating flat plate at a given span-wise position. For a non-zero V_∞ , the mean wake must transition from a bluff-body wake to a bound LEV topology as tip speed ratio increases, such that the full-saddle point, S, is replaced by a half-saddle, H. Note that the effective angle of attack, α , decreases with increasing tip speed ratio. A twisted plate will be compared to a flat plate in the experimental investigation in order to decouple the effects of tip speed ratio and angle of attack.

(non-zero $\tau(z)$) are compared in the current experimental investigation. The twist profile is such that the span-wise distribution of effective angle of attack is the same for the twisted plate and the flat plate at different values of tip speed ratio. In this way, the changes in the flow-field can be observed for different tip speed ratios at a constant distribution of effective angle of attack, and vice versa. Using particle image velocimetry (PIV) on span-normal planes, the flow topology will be qualitatively observed and circulation will be quantified.

The flux of vorticity through the leading-edge shear layer, J_s , requires a sink of vorticity to maintain a steady flow-field. Span-wise convection has been previously cited as the requisite sink (Ellington et al., 1996), although recent results show annihilation to be important in some cases (Wojcik and Buchholz, 2014). The velocity profile just ahead of the leading edge will be resolved using PIV on a small field of view (FoV), and the corresponding flux of vorticity through the leading-edge shear layer will be quantified. As the major source of

rotational fluid over the suction side of the plate, the trends of shear-layer vorticity flux could provide insight into the observed trends of steady-state circulation.

For each given test case, the span-wise variation of topology will also be investigated. It has been noted that the size of the region of rotational fluid in an LEV increases with span-wise position, as shown by Ellington et al. (1996) using smoke visualizations, by Garmann et al. (2013) using iso-surfaces of total pressure, and by Harbig et al. (2013) with iso-surfaces of Q-criterion. Correspondingly, the half-saddle behind a bound LEV must move towards the trailing edge with increasing span-wise position. The change in flow topology that occurs when the half-saddle meets the trailing edge will be investigated herein.

3.2 Experimental setup

An abstraction of natural samara flight was experimentally investigated in a free-surface water tunnel at the University of Calgary, for which more details are provided in Wong et al. (2013). A rotating plate was fixed along the axis of the tunnel, and subjected to a constant axial freestream velocity, V_∞ , as depicted in figure 3.2a. This flow configuration can be described with four parameters: tip speed ratio, twist profile, aspect ratio (AR), and Reynolds number (Re), defined as

$$AR = R/c, \text{ and} \\ Re = \frac{cv_{mid}}{\nu}, \quad (3.3)$$

where c is chord length, ν is kinematic viscosity and v_{mid} is the local effective velocity at the midspan, defined as

$$v_{mid} = \left(V_\infty^2 + (\Omega R/2)^2 \right)^{1/2}. \quad (3.4)$$

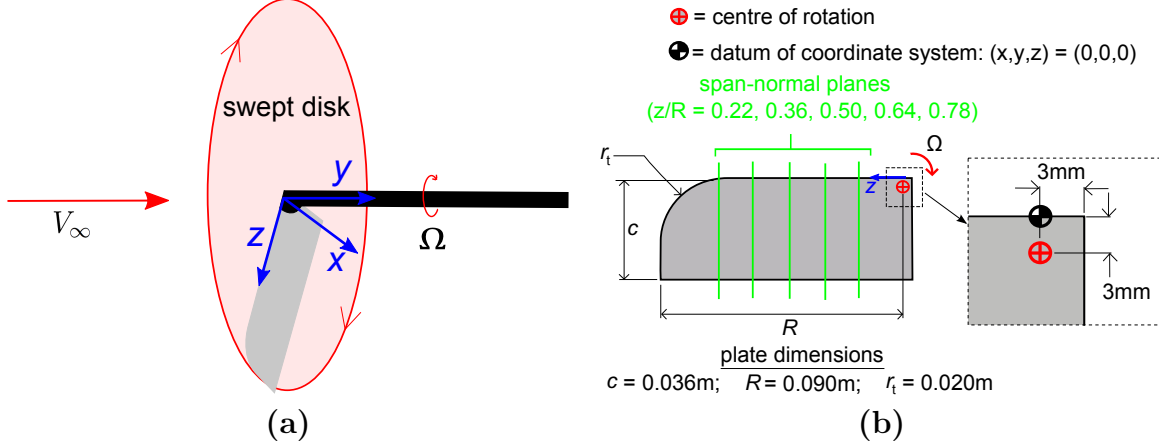


Figure 3.2: **(a)** Flow configuration for experiments. V_∞ and Ω are varied to achieve different tip speed ratios, λ . The plate-fixed coordinate system is centred at the leading edge in line with the root of rotation as shown here. **(b)** The dimensions of the chosen planform are shown here. Velocity fields on five span-normal planes were studied using particle image velocimetry (PIV).

A single planform was used for all test cases, such that AR was not varied. The chosen planform was a rectangular plate with a rounded corner at the tip of the leading edge. The chosen values of R , c , and radius of curvature at the tip, r_t , are given in figure 3.2b. The low AR and rounded tip are characteristic of insect wings and some species of samaras. Relevant studies in the literature use rectangular plates with square corners, as well as rounded, insect-inspired planforms.

Dye visualizations using potassium permanganate were completed at a constant freestream velocity of 0.016m/s while rotational speed was varied to achieve tip speed ratios of 0, 1.2, 2.4, and 3.6. Reynolds number, as defined in equation 3.3, varied with tip speed ratio, but within a narrow enough range so as to expect minimal differences ($Re \in [573, 1180]$). Two fixed geometric angles of attack were studied, $\tau(z) = 0^\circ$ and $\tau(z) = 20^\circ$, for all tip speed ratios. Figure 3.3a shows the experimental dye setup. Dye was injected into the flow from a hole on the pressure side of the plate near to the leading edge, at a span-wise location of $z/R = 0.28$. For each test, a pressure regulator was used to adjust the flow of dye until it was observed to barely exit the dye port and travel tightly around the leading edge. Dye was delivered to the pressure side port through the hollow shaft and hollow plate. A polymer

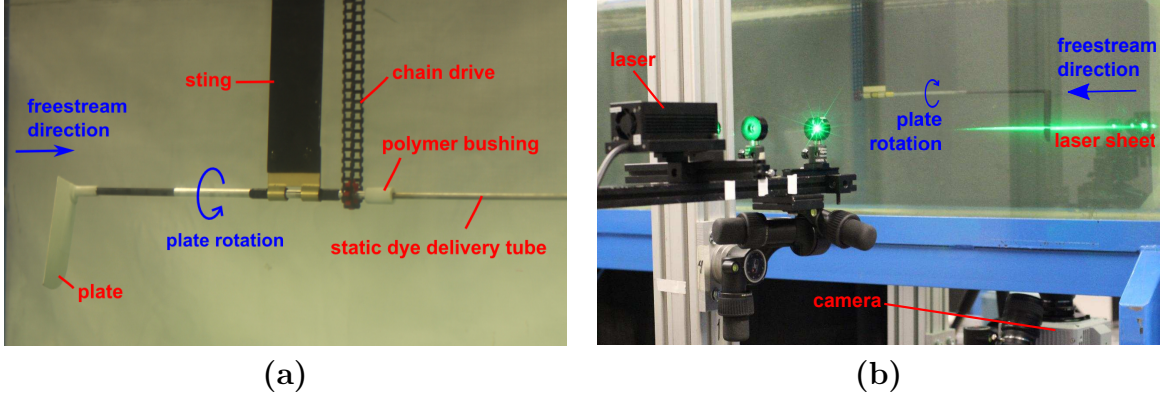


Figure 3.3: **(a)** Test apparatus set up for dye visualizations. Rotation of the plate is driven by a chain drive connected to an electric gearmotor above the waterline. **(b)** PIV setup in the water-tunnel. Two photos are taken once per rotation when the span-wise axis of the plate is perpendicular to the laser sheet. Five span-normal planes at different span-wise positions were sampled for each test case.

bushing was used to allow relative rotation of the static dye delivery tube and the rotating shaft connected to the plate. The rotation of the plate was driven via chain drive by an electric gearmotor above the surface of the water.

Planar PIV was performed with two different fields of view (FoVs) to quantify the flux of vorticity through the leading-edge shear layer (small FoV), and to view the overall flow topology and quantify mean circulation (large FoV). The width of these square FoVs were 0.0207m and 0.0964m, as shown in figure 3.4. The plate thickness was 1.57mm, or 0.044c. Three test cases were investigated: a flat plate was tested at tip speed ratios of 1 and 2.8, alongside a twisted plate at a tip speed ratio of 2.8. Twist profile is measured relative to the swept disk of the plate, and defined such that $\tau(z) \equiv 0$ for the flat plate. For the twisted plate, $\tau(z)$ was chosen such that at a tip speed ratio of 2.8, it has the same span-wise distribution of angle of attack as the flat plate at a tip speed ratio of 1. A photo of the flat and twisted plates is given in figure 3.5. The span-wise distributions of τ and α for the three test cases are shown in figures 3.6a and 3.6b, respectively.

Reynolds number was held constant at the midspan for each PIV test case: for the test case at a tip speed ratio of 1, the actual experimental value was $Re = 1250$, and for the

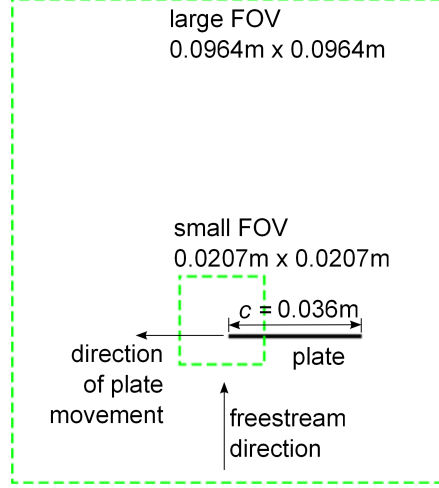


Figure 3.4: Using a 105mm lens, a small FoV was used to quantify shear-layer vorticity flux on span-normal planes. The experiment was then repeated with a 60mm lens to capture a larger FoV in order to determine the overall flow topology, and to quantify the mean circulation.

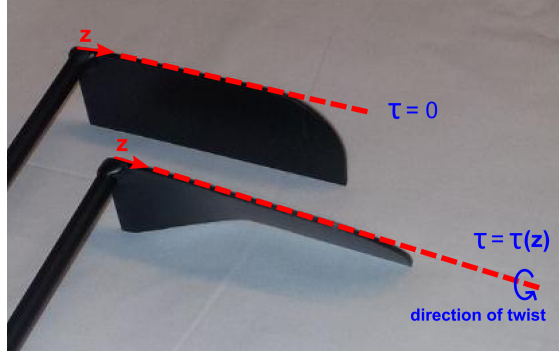


Figure 3.5: Photo of the twisted and flat plates. The twisted plate is twisted about the leading edge (span-wise axis) as shown. Twist profile $\tau(z)$ is chosen such that the α -distribution of the twisted plate at $\lambda = 2.8$ is equal to that of the flat plate at $\lambda = 1$ (see figure 3.6).

test cases at a tip speed ratio of 2.8 it was $Re = 1290$. For each of the test cases, five span-normal planes were sampled at span-wise positions of approximately $0.22R$, $0.36R$, $0.50R$, $0.64R$ and $0.78R$, as shown in figure 3.2b. Actual span-wise position varied slightly between test cases, and the measured span-wise position of each test plane is noted in the results. The experimental setup for PIV is very similar to that which was used for the dye visualizations, with the PIV hardware replacing the dye delivery system, as depicted in figure 3.3b. A Photron Fastcam SA4 camera was used with a 105mm and a 60mm lens to capture the small and large FoVs at frame rates of 500Hz and 125Hz, respectively. As

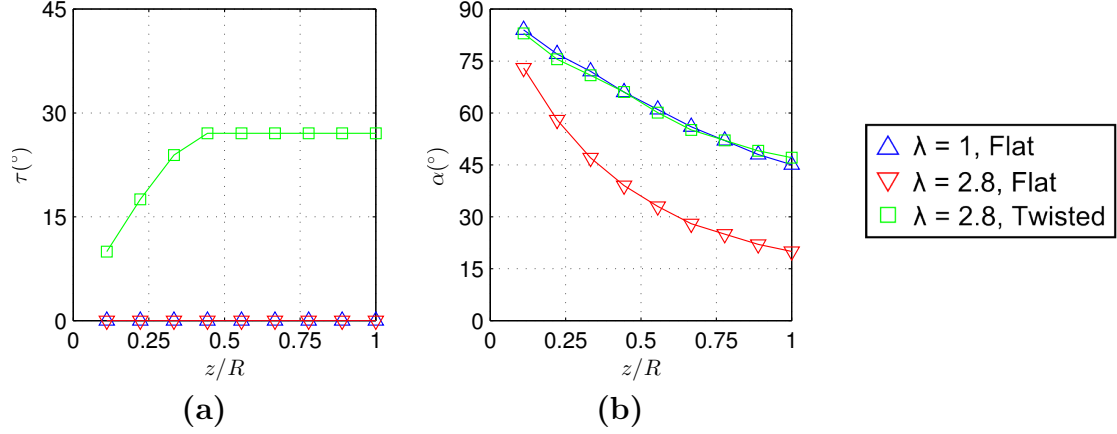


Figure 3.6: **(a)** Twist, τ , versus normalized span-wise position, z/R , for the three test cases. Twist is measured relative to the swept disk of the plate. **(b)** Local effective angle of attack, α , versus z/R for all three test cases. $\tau(z)$ for the twisted plate was chosen specifically to cause the shown collapse of its α -distribution to that of the flat plate at $\lambda = 1$.

the constantly rotating flat plate cut through the horizontal laser sheet, two consecutive images were captured when the span-normal plane was parallel to the laser sheet, yielding one correlation per rotation. Timing of the sampling was achieved with a phototransistor that changed state when the laser sheet was blocked by the passing of the plate. One hundred correlations per test plane were collected to allow the calculation of the mean velocity field.

Raw images from both cameras were captured at a resolution of 1024×1024 pixels, with a 12-bit depth. The spatial resolutions were $5.61 \times 10^{-4}c$ and $2.61 \times 10^{-3}c$ for the small and large FoVs, respectively. Multi-pass PIV correlations were performed: one pass with interrogation windows of 128×128 pixels and 75% overlap, and two passes with interrogation windows of 16×16 and 75% overlap. This processing produced one vector for every four pixels, which amounts to one vector every $2.24 \times 10^{-3}c$ for the small FoV, and every $0.0105c$ for the large FoV.

3.3 Results

3.3.1 Circulation on span-normal planes

The circulation associated with the leading-edge shear layer and LEV has been quantified, referred to hereafter as *leading-edge circulation* and denoted by Γ . In the chosen coordinate system, only negative-signed vorticity is therefore summed in the calculation of leading-edge circulation, which for convenience will be taken to be positive. Leading-edge circulation is normalized locally to give

$$\Gamma^* = \frac{\Gamma}{v_{\text{eff}}c}, \quad (3.5)$$

where v_{eff} is the local effective velocity, calculated as

$$v_{\text{eff}} = \left(V_{\infty}^2 + (\Omega z)^2 \right)^{1/2}. \quad (3.6)$$

Figure 3.7 shows the calculated values of Γ^* for each test plane plotted versus local effective angle of attack. Comparison of the twisted plate at a tip speed ratio of 2.8 and the flat plate at a tip speed ratio of 1 shows normalized leading-edge circulation to be independent of tip speed ratio at all span-wise locations. Circulation is uniquely defined when span-wise position, local effective angle of attack and local effective velocity are specified. At a constant tip speed ratio of 2.8, the increase in effective angle of attack associated with the twist profile of the plate causes an increase in circulation at all span-wise positions.

3.3.2 Flow topology

For a flat plate in a constant freestream velocity, four different tip speed ratios were investigated ($\lambda = 0, 1.2, 2.4$, and 3.6) using dye visualizations. The captured videos have been supplied as supplementary material. The dye is observed to trace a streakline around the edge of the recirculation zone for each case. As expected, and as described in section 3.1, the streakline draws closer to the suction side of the plate with increasing tip speed ratio

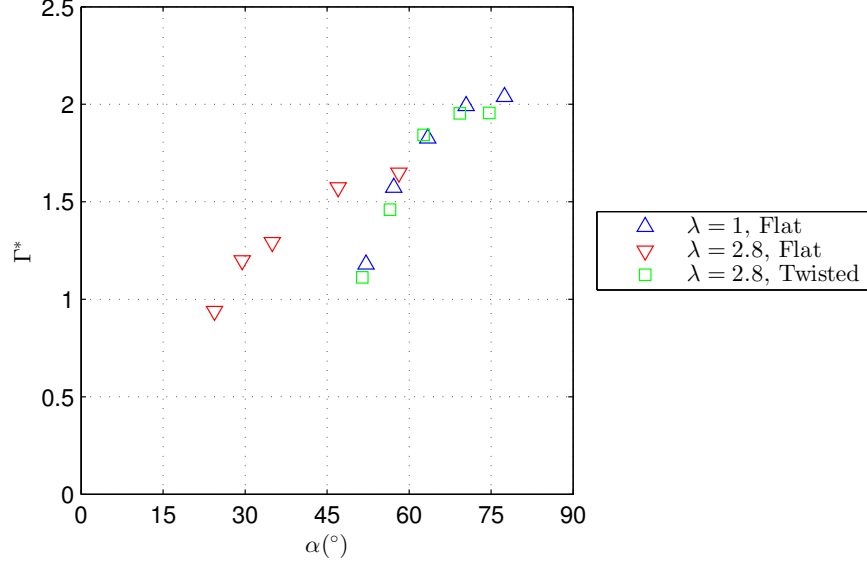


Figure 3.7: Normalized leading-edge circulation, Γ^* , versus local effective angle of attack, α . Trailing edge separation is not accounted for in the calculation of Γ^* . For each test case, each data point represents a different span-wise location. Γ^* collapses all along the span for the twisted plate at $\lambda = 2.8$ and the flat plate at $\lambda = 1$, showing Γ^* to be independent of λ . For constant $\lambda = 2.8$, an increase in α at each span-wise location causes an increase in Γ^* .

due to the draining of the recirculation zone by rotation-induced span-wise flow. For the flat plate at a tip speed ratio of 3.6, it is clear that the streakline stagnates onto the suction side of the plate, highlighting the existence of the bound-LEV topology described in section 3.1. For the inclined plate ($\tau = 20^\circ$) at a tip speed ratio of 3.6, the streakline spirals in a manner consistent with the presence of an LEV, and shows the tendency of the LEV to tilt away from the leading edge.

Contour plots of normalized span-wise vorticity, ω^* , from the mean velocity field of each test plane, are shown in figure 3.8, where ω^* is defined as,

$$\omega^* = \frac{\omega_z c}{v_{\text{eff}}}, \quad (3.7)$$

and ω_z is dimensional span-wise vorticity. One hundred PIV correlations (taken once per rotation) were averaged to obtain the mean velocity field on each test plane. The vorticity contours are overlaid with streamlines starting from the leading and trailing edges of the

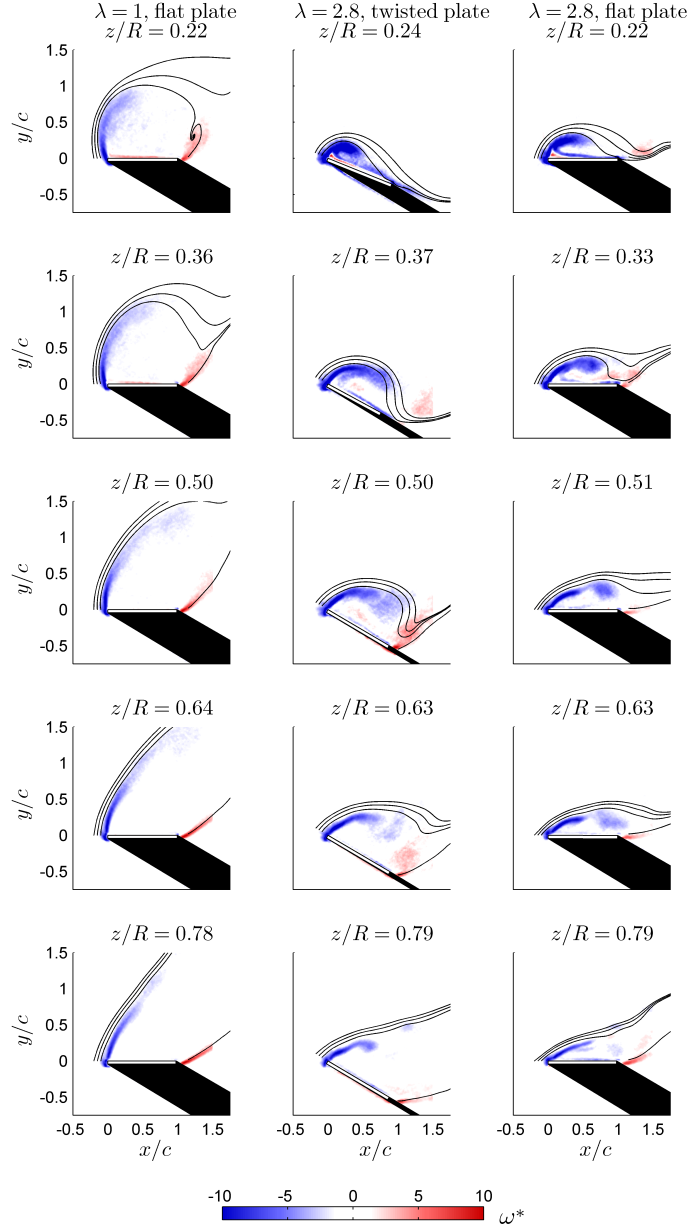


Figure 3.8: Contours of normalized span-wise vorticity, ω^* , from the mean velocity fields. Each column represents a different test case (different tip speed ratio and twist profile), as labelled at the top. Each row represents a different span-normal plane, moving in an outwards span-wise direction from top to bottom. A more compact recirculation zone is observed at inboard planes for higher λ , which is attributed to the increase in outwards span-wise flow. The white rectangle in each image represents the plate, and the black regions are in shadow. The plate moves from right to left through the FoV, and the freestream flow is from the bottom to the top of the FoV.

plate, as viewed from a plate-fixed frame of reference. At a tip speed ratio of 1, the flow separates from both the leading and trailing edges at all examined test planes. The mean velocity field resembles a bluff-body wake, featuring a full-saddle point above the suction side of the plate. At a tip speed ratio of 2.8, a bound LEV topology is observed at the most inboard planes. Moving outwards from the centre of rotation, the half-saddle behind the LEV moves towards the trailing edge. Outboard of the critical span-wise location where the saddle point reaches the trailing edge, separation at the trailing edge becomes obvious, and the half-saddle is replaced by a full-saddle in the wake. For different angle-of-attack distributions at equal tip speed ratio, the flow topology was not greatly changed, as seen by comparing the two far-right columns in figure 3.8. For outboard planes, where the flow-field resembles that of a bluff-body wake, the angles of the leading- and trailing-edge shear layers appear to be determined by local effective angle of attack, but tip speed ratio appears to have a greater effect on where a bound LEV is observed.

3.3.3 Shear-layer measurements

The shear-layer velocity profile at the leading edge of the plate was quantified using PIV with a small FoV. The velocity was sampled along a line extending from the suction side of the plate for each test case (along the ξ coordinate), as shown in figure 3.9. One hundred correlations were averaged to produce the mean velocity profiles, which were then normalized by average effective velocity, $\overline{v_{\text{eff}}}$, defined as

$$\overline{v_{\text{eff}}} = \frac{1}{R} \int_0^R v_{\text{eff}}(z) dz. \quad (3.8)$$

The resulting normalized velocity profiles are plotted in figure 3.10.

A key challenge with PIV is resolving velocity vectors close to a body due to reflections. In the current experiment, it is estimated that a saturated bright spot ahead of the leading edge extends, at worst, one-third of the plate thickness ahead of the leading edge; as such,

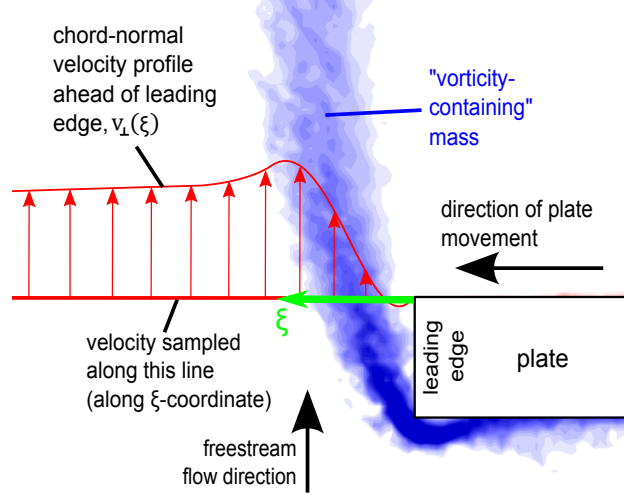


Figure 3.9: Chord-normal velocity is sampled along a horizontal line extending from the leading edge of the suction side of the plate. Vorticity flux through the shear layer, J_s , is the integral across the shear layer of chord-normal velocity times span-wise vorticity (see equation 3.9). Vorticity contours are shown in blue to highlight the rotational fluid in the shear layer.

the six innermost vectors (from $\xi/c = 0$ to about $\xi/c = 0.015$) are suspect, although the value of velocity tends to zero as expected by the no-slip condition. The remainder of the velocity profile is unaffected by the reflections at the plate.

The maximum velocity in the shear layer, $v_{\max}$, is fairly constant along the span for all test cases. Furthermore, when normalized by $\overline{v_{\text{eff}}}$, the maximum shear-layer velocity is constant for all test cases ($v_{\max}/\overline{v_{\text{eff}}} \approx 1.3$). The vorticity flux through the shear layer, J_s , was also relatively insensitive to span-wise position. J_s is calculated as

$$J_s = \int_{SL} v_{\perp} \omega_z d\xi, \quad (3.9)$$

where $v_{\perp}$ is the chord-normal velocity ahead of the leading edge, and SL denotes the shear layer. The integration is taken from the leading edge ($\xi = 0$) to a point beyond the shear layer where vorticity approaches zero. A similar approach has been previously used to successfully predict circulation growth for two-dimensional problems (Sattari et al., 2012; Wong et al., 2013), and was used by Wojcik and Buchholz (2014) in quantifying the balance of circulation

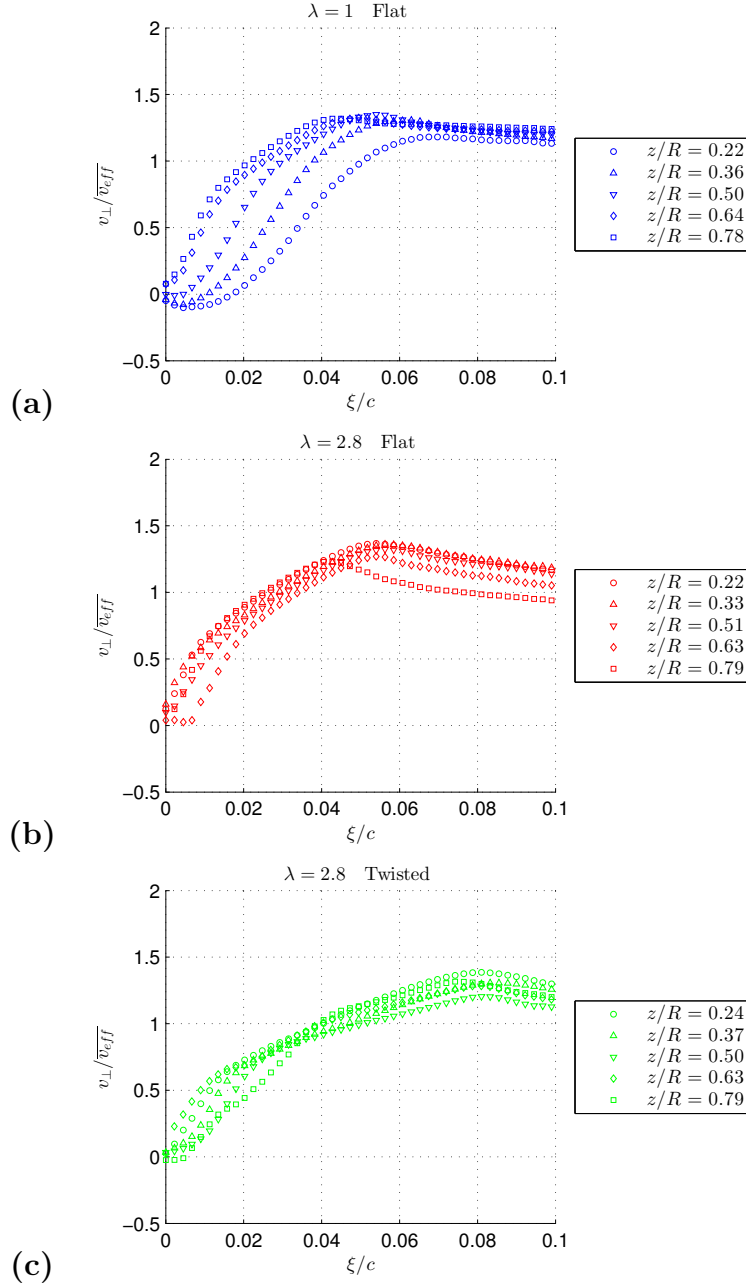


Figure 3.10: Normalized shear-layer velocity profiles, $v_{\perp}/\overline{v_{\text{eff}}}$ for **(a)** the flat plate at $\lambda = 1$, **(b)** the flat plate at $\lambda = 2.8$, and **(c)** the twisted plate at $\lambda = 2.8$. Velocity is sampled along a line colinear with the suction side of the plate on each test plane (see figure 3.9), as denoted by the coordinate ξ . Normalized peak velocity is relatively flat along the span for each test case, and is equal to about 1.3 for all test cases. Shear-layer thickness is greatest for the twisted plate at $\lambda = 2.8$, leading to greater shear-layer vorticity flux.

transport over a rotating flat plate. Normalized leading-edge vorticity flux, J_s^{**} , is defined as

$$J_s^{**} = \frac{J_s}{v_{\text{eff}}^2}. \quad (3.10)$$

The superscript ** has been used to denote normalization by a global parameter, which is constant for a given test case across the entire span. Previously, the superscript * was used to denote normalization by local, in-plane parameters, which vary with span-wise location. An increase in tip speed ratio for a constant angle-of-attack distribution gives rise to an increase in normalized shear-layer flux, as shown in figure 3.11a. An increase in effective angle of attack at constant tip speed ratio has the same effect. Conspicuously, normalized shear-layer flux collapses all along the span for the flat plate test cases, which share neither a common tip speed ratio nor distribution of effective angle of attack. Unless the competing effects of tip speed ratio and angle of attack coincidentally cancel for the two flat plate cases, it would appear that the twist profile of the plate, $\tau(z)$, is the governing parameter of J_s^{**} . As such, the shear-layer vorticity flux and the local circulation are driven by different physical parameters, and are not correlated to one another. This is shown by defining a normalized ratio of shear-layer vorticity flux to circulation,

$$\frac{J_s^{**}}{\Gamma^{**}} = \frac{J_s c}{\Gamma v_{\text{eff}}}. \quad (3.11)$$

This ratio is plotted versus span-wise position in figure 3.11b. Although the ratio is similar on the inboard planes at a constant tip speed ratio of 2.8, no clear trend in the data is observed; equilibrium leading-edge circulation is not directly correlated to shear-layer vorticity flux.

3.4 Discussion

The observed independence of circulation from tip speed ratio implies that the equilibrium is driven by the local inflow conditions of effective angle of attack and effective velocity,

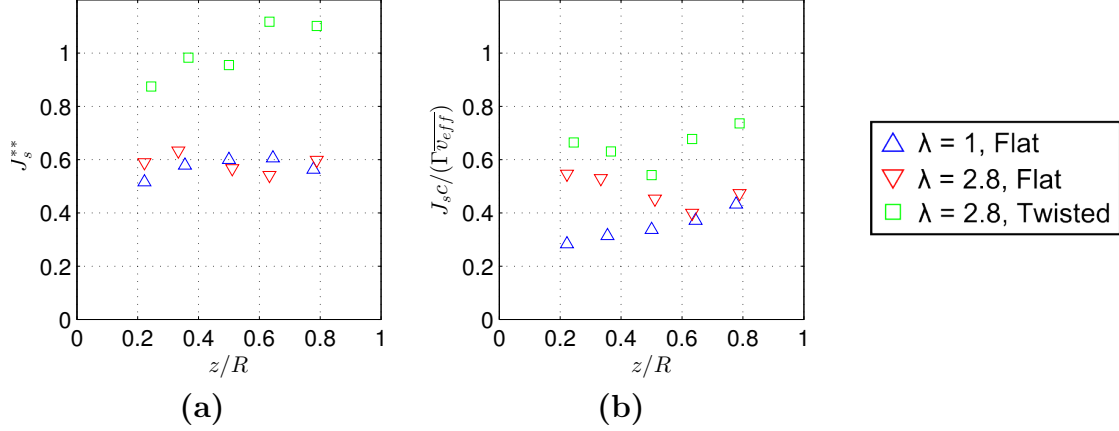


Figure 3.11: **(a)** Normalized vorticity flux through the leading-edge shear layer, J_s^{**} , versus normalized span-wise position, z/R . Twist profile appears to govern shear-layer vorticity flux, as the data for both flat plate test cases collapse independently of tip speed ratio. **(b)** Normalized ratio of shear-layer flux to circulation, plotted versus z/R . No obvious collapse is observed, showing that there is no clear correlation between shear-layer vorticity flux and steady-state circulation.

and modulated by span-wise position. This is analogous to thin aerofoil theory, wherein the topology of the flow is fixed by specifying a Kutta condition at the trailing edge, and by determining the pressure-side stagnation point based on angle of attack. Variation of local effective velocity then serves to scale the flow field and the circulation accordingly. For a given angle of attack in a two-dimensional flow around an aerofoil, normalized circulation is thus fixed. For a finite wing, the proximity of the tip vortex is added as an additional degree of freedom. For a finite, rotating plate undergoing separation, yet another degree of freedom exists; the fixed saddle point given by the Kutta condition is replaced by a full-saddle point in the wake when both leading- and trailing-edge separation occur, or a near-Kutta condition exists in conjunction with a half-saddle on the suction side of the plate for the bound-LEV topology. Tip speed ratio was shown to be critical in determining the location of the saddle point (see figure 3.8), but does not correspondingly alter leading-edge circulation. The net effect of increased tip speed ratio can thus be interpreted as increased vortex stretching, whereby the vorticity distribution around the leading edge is made more compact, but the total leading-edge circulation remains unchanged.

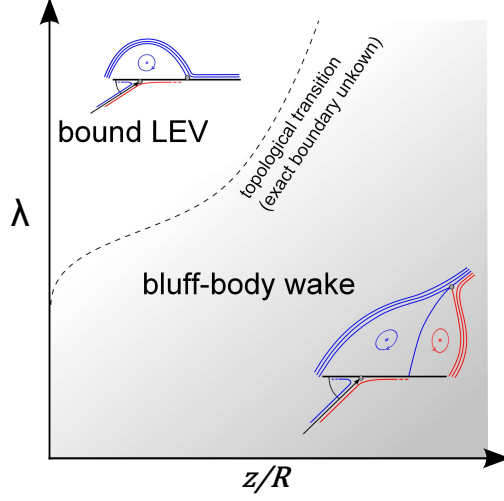


Figure 3.12: Topological transition in the two-dimensional parameter space of tip speed ratio, λ , and normalized span-wise position, z/R . A minimum tip speed ratio is necessary to generate the bound LEV topology, which is then observed only at span-wise positions sufficiently close to the centre of rotation. In zero freestream conditions, where tip speed ratio approaches infinity, the transition is expected to approach a maximum span-wise position for a given planform. Effective angle of attack appears to have a lesser effect on the span-wise location of the topological transition than tip speed ratio.

In previous studies of rotating plates in zero freestream conditions, it was shown that an LEV is most likely to be observed at inboard locations, or areas with low local Rossby number (Lentink and Dickinson, 2009). With the addition of a freestream velocity parallel to the axis of rotation, a two-dimensional parameter space arises, wherein the bound-LEV topology is possible at sufficiently high tip speed ratios, and at span-wise positions sufficiently close to the centre of rotation (see figure 3.12). Tip speed ratio tends to infinity for zero freestream conditions (i.e. propeller hover conditions), and it is expected that the span-wise position of the topological transition approaches some maximum for a given planform. Effective angle of attack may affect this transition to some degree, although the variation of angle-of-attack distribution at constant tip speed ratio appears to have only a secondary impact on topology in the current investigation.

The flux of vorticity through the leading-edge shear layer requires a sink of vorticity in order to maintain the steady flow-field characteristic of stable LEVs. Circulation balance

is considered using an integral form of the vorticity transport equation, shown here as a modified version of the equation presented by Wojcik and Buchholz (2014):

$$\frac{\partial \Gamma}{\partial t} = \int_B -w \frac{\partial \omega_z}{\partial z} dB + \int_B \left((\omega_y + 2\Omega) \frac{\partial w}{\partial y} + \omega_x \frac{\partial w}{\partial x} \right) dB + \int_B \omega_z \frac{\partial w}{\partial z} dB + J_s - \Phi, \quad (3.12)$$

where w is span-wise velocity and ω is vorticity, with the subscripts denoting orientation. Ω enters the equation in the second integral as the curl of Coriolis acceleration. Φ is vorticity annihilation, shown by Wojcik and Buchholz (2014) to be important to the overall circulation balance in some cases. B is an area on a span-normal plane that encompasses all of the rotational fluid. The integrals on the right-hand side represent span-wise convection, vortex tilting, and vortex stretching, as read in order from left to right.

For a stationary plate, no bulk span-wise flow exists to convect vorticity in a span-wise direction, and the shear layer flux must be balanced by in-plane convection (i.e. vortex shedding) and diffusion. As rotation is increased, span-wise convection increases, and a steady-flow field becomes possible at some critical tip speed ratio. Rotation will cause span-wise acceleration of fluid, such that positive $\partial w / \partial z$ is expected in the vicinity of the rotational fluid, and vortex stretching occurs. However, vortex stretching serves only to redistribute the vorticity above the plate without altering the total circulation. Tilting would increase, as the Coriolis acceleration acting on the span-wise flow in the recirculation zone will increase. Although Wojcik and Buchholz (2014) showed tilting to have no net effect on circulation transport in their experiment at locations inboard of the midspan, it is expected to play a role at locations approaching the tip where any stable vortex system must begin to tilt into the helical wake. Annihilation can also be expected to increase, as the distribution of vorticity becomes more compact near the plate. However, the magnitudes of each of these transport mechanisms do not necessarily scale equivalently with increased tip speed ratio, such that the independence of normalized circulation from tip speed ratio is not directly predicted from this sort of inspection of the governing equation. If the relative magnitudes

of all of these mechanisms were insensitive to tip speed ratio, then one might expect shear-layer vorticity flux to be consistently correlated to local circulation for a constant tip speed ratio. However, figure 3.11b shows a normalized ratio of shear-layer vorticity flux to circulation, and no clear collapse of the data is observed for any of the three different test cases. Instead, changes in shear-layer vorticity flux do not affect circulation at all if the angle-of-attack distribution remains constant. It would appear that the remaining vorticity transport mechanisms (convection, tilting and annihilation) adjust to such changes in vorticity flux, conspiring to maintain the equilibrium circulation established by the span-wise position and the local, irrotational inflow conditions.

Another approach to explaining the equilibrium of circulation may be warranted. Zannetti and Gourjii (2014) investigated the equilibrium positions of counter-rotating vortex pairs in the wake of an infinite flat plate, where the strength of the vortices and the total global circulation determine the topology of the wake. For any angle of attack, they showed that the total global circulation, encompassing both the leading- and trailing-edge circulation, must be equal to zero. Zero global circulation ensures that the wake is closed, and that a local maximum of vorticity does not exist away from the plate in violation of the two-dimensional maximum principle for vorticity (Lugt, 1985). For a steady three-dimensional problem, this constraint does not apply, as local maxima of vorticity are permitted in regions where diffusion is balanced by vortex stretching. However, the observations herein tend to imply that some constraint on leading-edge circulation does apply, and could be similarly enforced by the allowable topologies of the irrotational flow around the recirculation region. Although not trivial, an extension of this type of wake analysis to three dimensions could provide insight into the problem.

For a discussion of force production, the total circulation around the plate is relevant. Trailing-edge circulation was not quantified, but the observed variation in trailing-edge separation suggests that it is not fixed by local inflow conditions like leading-edge circulation,

but rather is altered by tip speed ratio along with the topological changes. For translating plates, Pitt Ford and Babinsky (2013) showed bound circulation to be small after the initial development of an LEV, and it was argued that this result would be valid for rotating plates shortly after the onset of motion. For steadily-rotating plates, after the three-dimensional flow-field has had time to fully develop, the expected bound circulation is unclear. In spite of these unknowns, some preliminary comments on the effect of tip speed ratio on force production can be made through consideration of leading-edge circulation alone. The bound-LEV topology is expected to be the most advantageous in terms of force production, as the contraction of the leading-edge vorticity nearer to the plate at higher tip speed ratio is expected to increase the suction peak acting on the plate. The sharpness of the leading edge corners prevents leading-edge suction, such that the enhanced force is still expected to be normal to the plate for the range of Reynolds number investigated here (Birch et al., 2004). Outboard of the midspan, the topology more closely resembles a bluff-body wake in the current investigation, as noted in section 3.3.2. The redistribution of the vorticity into trailing shear layers at outboard locations is expected to decrease force production, presumably approaching two-dimensional, flat-plate sectional force coefficients.

3.5 Conclusions

Flow topology and leading-edge circulation over steadily rotating plates in a freestream flow were studied using planar PIV. The effects of tip speed ratio and effective angle of attack were independently observed by comparing plates with different twist profiles. Circulation on span-normal planes is uniquely defined by span-wise position and local inflow conditions (local effective angle of attack and local effective velocity). Changes in tip speed ratio do not alter this equilibrium circulation when the inflow conditions are held constant. The shear-layer velocity profiles ahead of the leading edge were resolved using a sufficiently small field of view. This allowed shear-layer vorticity flux to be quantified, which was shown

to be uncorrelated to the observed equilibrium values of circulation. This insensitivity of circulation to the leading-edge vorticity flux suggests that the remaining vorticity transport mechanisms (span-wise convection, tilting, annihilation) tend to adjust in magnitude to maintain the equilibrium circulation set by the irrotational inflow conditions.

Tip speed ratio is the key driver of the topology of the flow, where an increase in tip speed ratio stretches the leading-edge vorticity and drains the recirculation zone to make it more compact. At sufficiently high tip speed ratio, the flow that separates from the leading edge stagnates onto the suction side of the plate and reattaches, encompassing a bound LEV. This is observed at inboard locations; moving outwards from the centre of rotation, the stagnation point behind the LEV moves towards the trailing edge. At sufficiently outboard locations, the half-saddle on the plate is replaced by a full-saddle in the wake, and separation at the trailing edge occurs. Inboard of the critical location where the saddle point meets the trailing edge, the bound LEV is expected to enhance force production. Tip speed ratio may serve as a good predictor of the span-wise location separating these distinct topologies, while local effective angle of attack, local effective velocity and span-wise position may be used to independently predict leading-edge circulation.

Acknowledgements

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Chapter 4

Robust gust-response of autorotating samaras

Abstract

Samara-seeds exhibit robust autorotation, as shown here with high-speed video of an *Acer negundo* samara hitting a sharp-edged gust. Despite a rapid change in trajectory, the samara is observed to maintain autorotation as it descends through the horizontal flow generated by an electric fan. Momentum theory is used to predict the time constant associated with the post-gust acceleration of the samara. The theoretical inertial time constant overpredicts the experimentally obtained value by 20%, demonstrating the derived momentum model to be a valid approximation. The model suggests that increasing gust frequency or gust strength will generate greater transient changes in tip speed ratio. Sufficiently large perturbations to the equilibrium tip speed ratio can compromise the pitch-stability of an autorotating samara.

4.1 Introduction

The term *samaras* refers to seeds that tend to rotate about a vertical axis as they descend from their parent plant. This kinematic pattern will be referred to as *autorotation* in the present work, although by some classical definitions it may be considered *pseudo-autorotation* (Lugt, 1983). The generation of lift from this flight pattern slows the descent of the seed in order for it to be carried further by horizontal winds (Greene and Johnson, 1990). Achieving sufficient dispersal distance is crucial to avoiding sibling competition, and to improving the probability that one or more samaras will land in safe sites to germinate.

The energy required to drive the continuous rotation of a samara comes from the conversion of gravitational potential energy as the samara descends. Without an external source of

energy, samaras cannot hover; they must descend at a non-zero velocity. Azuma and Yasuda (1989) derived an expression for the minimum possible descent velocity by treating a samara as an actuator disk and applying momentum theory. Herein, a samara that achieves this minimum descent velocity will be referred to as *optimal*, and its flight will be referred to as *optimal autorotation*. Azuma and Yasuda (1989) found that for a range of disk-loading values, the flight of several species of samaras tends to approach optimal autorotation, as one might expect through natural selection. Interestingly, samara flight has evolved multiple times within the plant kingdom (Matlack, 1987), highlighting its flight as a particularly successful reproductive strategy.

In order for a samara to experience the benefit of reduced descent velocity, its autorotative kinematics must be somewhat robust to disturbances. If autorotation is disrupted by a gust, a samara will fall rapidly for a period of time until it can re-establish autorotation. Therefore, natural selection has likely favoured flight patterns that are not only slow in descent, but also robust enough to navigate gusty and turbulent environments. The gust-response of an *Acer negundo* samara will be investigated in the present work using high-speed video. Its tendency to maintain autorotation through a gust is achieved passively, as samaras possess no control authority. The mechanisms of passive stability could provide insight into the design of bio-inspired aerial vehicles.

Norberg (1973) considered the dynamics of falling samaras, and the conditions necessary to maintain stable autorotation. In order to maintain a dynamically stable pitch angle, the chord-wise centre of mass (CoM) must lie near the one-third chord location from the leading edge where it can align with the chord-wise centre of pressure (CoP) (Norberg, 1973). When the flow around the samara is separated, an increase in pitch angle causes the CoP to move towards the trailing edge. This CoP movement creates a restoring moment to reduce the pitch angle, and the reverse happens with a decrease in pitch angle (see figure 4.1).

More recently, stable leading-edge vortices (LEVs) have been shown to be a feature in the

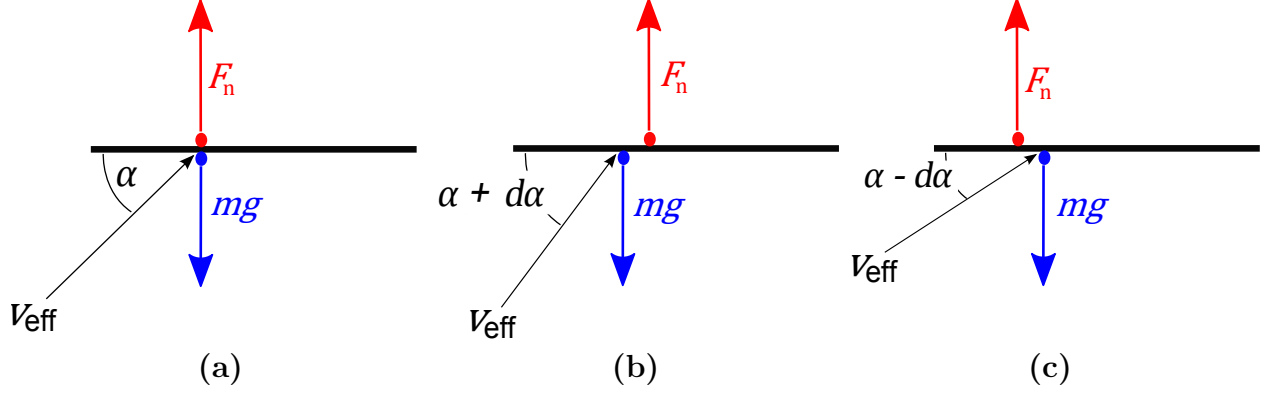


Figure 4.1: **(a)** Mechanism of pitch-stability in samara flight based on the alignment of the weight, mg , and the aerodynamic force, F_n , as proposed by Norberg (1973). The diagram shows a cross-section of a samara on a span-normal plane. **(b)** An increase in pitch angle causes the centre of pressure to move towards the trailing edge, creating a restoring moment to reduce the pitch angle. **(c)** A decrease in pitch angle has the opposite effect, causing a restoring moment in the reverse direction. (Graphic adapted from Norberg (1973))

flight of several species of samaras (Lentink et al., 2009; Lee et al., 2014). In this context, *stable* refers to the position of the LEV core, which maintains a mean convection speed of zero in the samara-fixed frame of reference. A cartoon of the flow-field, as investigated previously by Lentink et al. (2009) and Lee et al. (2014), is given in figure 4.2a. Figure 4.2b shows a cross-section of the flow-field on a span-normal plane. The presence of a stable LEV enhances force production (Lentink et al., 2009), and may serve a further purpose in samara flight. The velocity field around an LEV is relatively steady compared to the conditions of bluff-body vortex shedding; the instantaneous CoP is therefore expected to vary less from its mean location, which may improve the robustness of autorotation by preventing pitch-instability.

Tip speed ratio, λ , was shown in chapter 3 to be an important parameter for stable LEV attachment in steady-state rotation. Stable LEVs were qualitatively shown to exist at tip speed ratios of 2.4 and 3.6, but were not definitively identified at a tip speed ratio of 1.2. A review of the literature on various species of samaras shows equilibrium tip speed ratios between 1.7 and 3.6 (see table 4.1), suggesting that many of these samaras likely benefit from stable LEV formation. The Reynolds numbers for all of the listed species is of the

order $O(10^2)$ to $O(10^3)$. Gusts parallel to the axis of rotation can alter the instantaneous tip speed ratio, which could initiate LEV detachment. Changes in tip speed ratio are also accompanied by changes in effective angle of attack, thereby moving the CoP location and possibly compromising pitch-stability. The tendency of gusts to perturb the tip speed ratio from equilibrium will be considered herein. Momentum theory will be applied to describe the response of a samara to a gust. The resulting model will be validated by tracking the post-gust displacement of an *Acer negundo* samara using high-speed video, as described in greater detail in section 4.3.

4.2 Theoretical model

The response of a samara to an axial gust is treated theoretically in this section. The term *axial* will be used throughout this document to refer to the direction parallel to the axis of rotation, whereas *transverse* will refer to the perpendicular direction. A schematic of an autorotating samara encountering an axial gust is depicted in figure 4.3.

A samara dropped from a sufficient height will enter autorotation and establish equilibrium kinematics, reaching constant values of descent velocity, v_d , rotational speed, Ω , and coning angle, β (measured relative to the swept disk). The axial gust will be treated as a change in the axial freestream velocity, V_∞ . A gust causes a change in the tip speed ratio, disturbing it from equilibrium according to the relation

$$\lambda = \frac{\Omega R_d}{v_d + V_\infty}, \quad (4.1)$$

where R_d is the radius of the swept disk. The flow-field as viewed from a samara-fixed frame of reference is dependent on the equilibrium tip speed ratio, as shown in chapter 3 (see figure 3.8). Significant departure from this equilibrium could disrupt the stability of autorotation by altering the CoP location. The inertia of the samara will determine how much the tip speed ratio is disturbed, and how quickly it returns to equilibrium after encountering a gust.

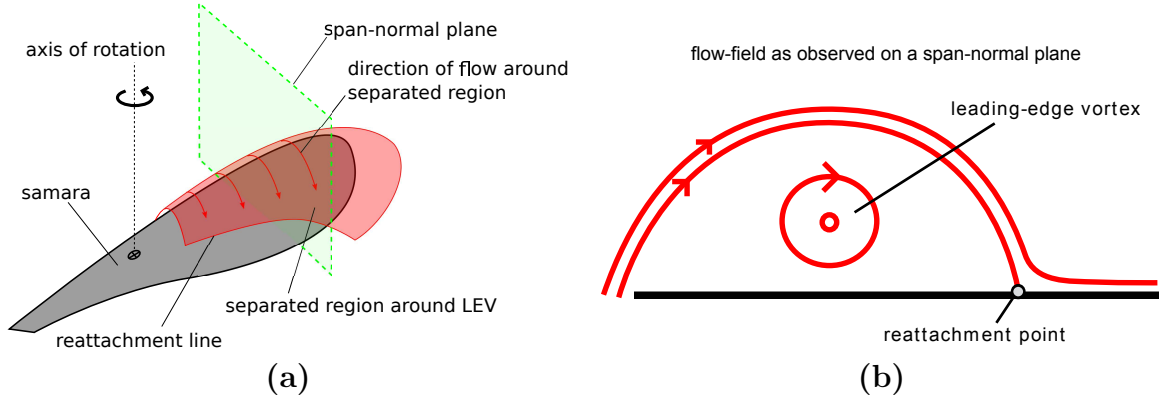


Figure 4.2: **(a)** Cartoon of the three-dimensional flow-field around an autorotating samara, as investigated previously by Lentink et al. (2009) and Lee et al. (2014). The presence of a stably-attached LEV enhances force production, and is hypothesized herein to stabilize the instantaneous CoP location to improve pitch-stability. **(b)** The flow-field as observed on a span-normal plane at a sufficiently inboard location. The LEV core does not convect into the wake, but remains in place relative to the samara.

Species	λ	Source
<i>Acer platanoides</i>	3.36	Norberg (1973)
<i>Picea abies</i>	2.16	Norberg (1973)
<i>Acer diabolicum</i> Blume	3.54	Azuma and Yasuda (1989)
<i>Acer palmatum</i> Thunb.	2.08	Azuma and Yasuda (1989)
<i>Acer palmatum</i> Thunb. Matsumurae Makino	2.32	Azuma and Yasuda (1989)
<i>Carpinus Tschonoskii</i> Maxim	1.74	Azuma and Yasuda (1989)
<i>Pinus Thunbergii</i> Parlatores	2.82	Azuma and Yasuda (1989)
<i>Cedrus deodora</i>	2.99	Minami and Azuma (2003)
<i>Pinus luchuensis</i>	2.59	Minami and Azuma (2003)
<i>Acer palmatum</i>	2.06	Lee et al. (2014)

Table 4.1: Equilibrium tip speed ratios of various samara species, as extracted from the literature. The observed natural range of tip speed ratio is similar to that shown in chapter 3 to allow for stable LEV attachment in steady-state.

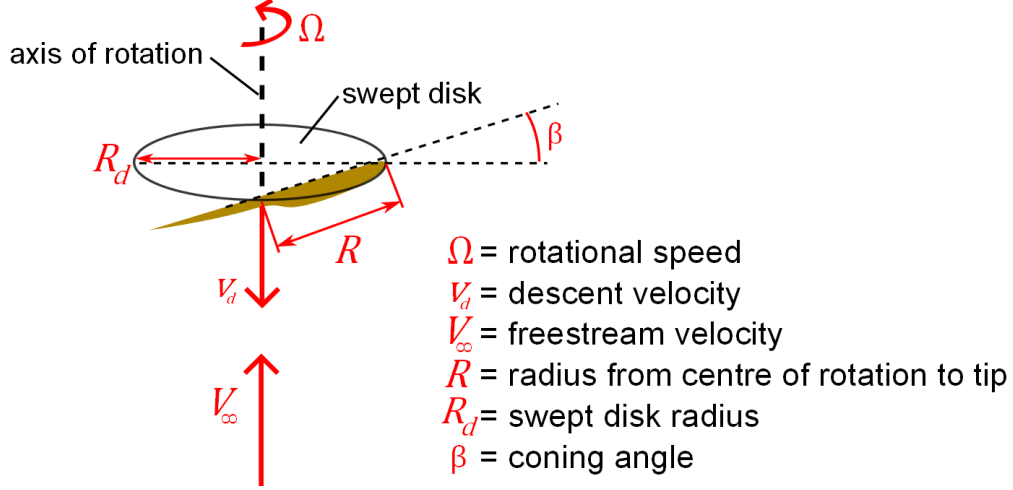


Figure 4.3: Schematic of an autorotating samara. To model an axial gust, the freestream velocity, V_∞ , will be treated as an input to the system. The time constant of acceleration associated with the gust-response will be derived using momentum theory.

To model the inertial response, we will treat an autorotating samara as an actuator disk and apply momentum theory, as done previously by Azuma and Yasuda (1989). Consider the force balance acting on a descending samara:

$$m\dot{v}_d = mg - T, \quad (4.2)$$

where m is seed mass, g is the acceleration due to gravity, and T is the upwards thrust acting on the samara due to the downward acceleration of fluid around it. Descent velocity is defined as positive in the downwards direction, whereas the freestream velocity is defined as positive upwards, as portrayed in figure 4.3. Since the incident gust disturbs the equilibrium kinematics, we begin by considering the actuator disk to have dynamic inflow, as developed by Carpenter and Fridovich (1953). Their expression for upwards thrust on a rotor experiencing dynamic inflow is given below, adapted to the nomenclature of the current work:

$$T = m_a \dot{v}_i + 2\rho S(v_r - v_i - \frac{2}{3}\dot{\beta}R_d)v_i, \quad (4.3)$$

where m_a is the apparent mass of fluid displaced by the accelerating body, v_i is the induced

velocity (assumed constant over the disk), v_r is the axial velocity of the fluid relative to the swept disk, ρ is the density of the surrounding fluid, S is the area of the swept disk, and $\dot{\beta}$ is the rate of change of the coning angle. The induced velocity is assumed to be constant through a gust, eliminating the apparent mass term and preserving the linearity of the thrust term in equation 4.2. In cases where the inertial response is slow, and the flow-field relative to the samara is significantly disturbed, this assumption becomes less valid. The change in coning angle, and its rate of change, will be assumed to be small; the validity of these assumptions will be assessed qualitatively using high-speed video. Given these assumptions, equation 4.3 reduces to the steady-state equation for thrust acting on an actuator disk:

$$T = 2\rho S(v_r - v_i)v_i. \quad (4.4)$$

The velocity of fluid relative to the swept disk can be broken into two components,

$$v_r = v_d + V_\infty. \quad (4.5)$$

For samaras that approach optimal autorotation, i.e. those that achieve the minimum possible descent velocity, the induced velocity is equal to half of the initial descent velocity, v_0 , as presented by Azuma and Yasuda (1989):

$$v_i = \frac{1}{2}v_0. \quad (4.6)$$

The samara in question will be assumed to operate in a near-optimal fashion, which will be validated experimentally. The resulting model is

$$\dot{v}_d + \left(\frac{\rho S v_0}{m}\right)v_d = g + \left(\frac{\rho S v_0^2}{2m}\right) - \left(\frac{\rho S v_0}{m}\right)V_\infty. \quad (4.7)$$

The time constant of this first order linear system, τ_i , is the reciprocal of the coefficient on the v_d term:

$$\tau_i = \frac{m}{\rho S v_0}. \quad (4.8)$$

Expressing the model in terms of this time constant, we arrive at

$$\tau_i \dot{v}_d + v_d = \tau_i g + \frac{1}{2} v_0 - V_\infty. \quad (4.9)$$

Having assumed near-optimal autorotation, it can be shown that the equilibrium descent velocity is given by

$$v_0 = 2\tau_i g, \quad (4.10)$$

and the model can be expressed as

$$\tau_i \dot{v}_d + v_d = v_0 - V_\infty. \quad (4.11)$$

In the experimental investigation, a sharp-edged gust is generated and treated as a step-change in disk-normal velocity. Such an assumption allows for the extraction of an experimental time constant to validate the expression in equation 4.8. The discussion in section 4.5 considers sinusoidal gusts of varying period, which is more reflective of the gusts encountered in an outdoor environment.

4.3 Methods

High-speed video of an autorotating *Acer negundo* samara was captured using a Fastcam Photron SA4 camera operating at 2000 frames per second. The samara was observed as it fell in quiescent air to determine its equilibrium kinematics. Its response to a gust was then captured by dropping the samara in front of an electric fan, simulating a sharp-edged, transverse gust. The test specimen was dropped from 1.2m above the fan to allow it to reach equilibrium autorotation prior to entering the field of view. The samara descended

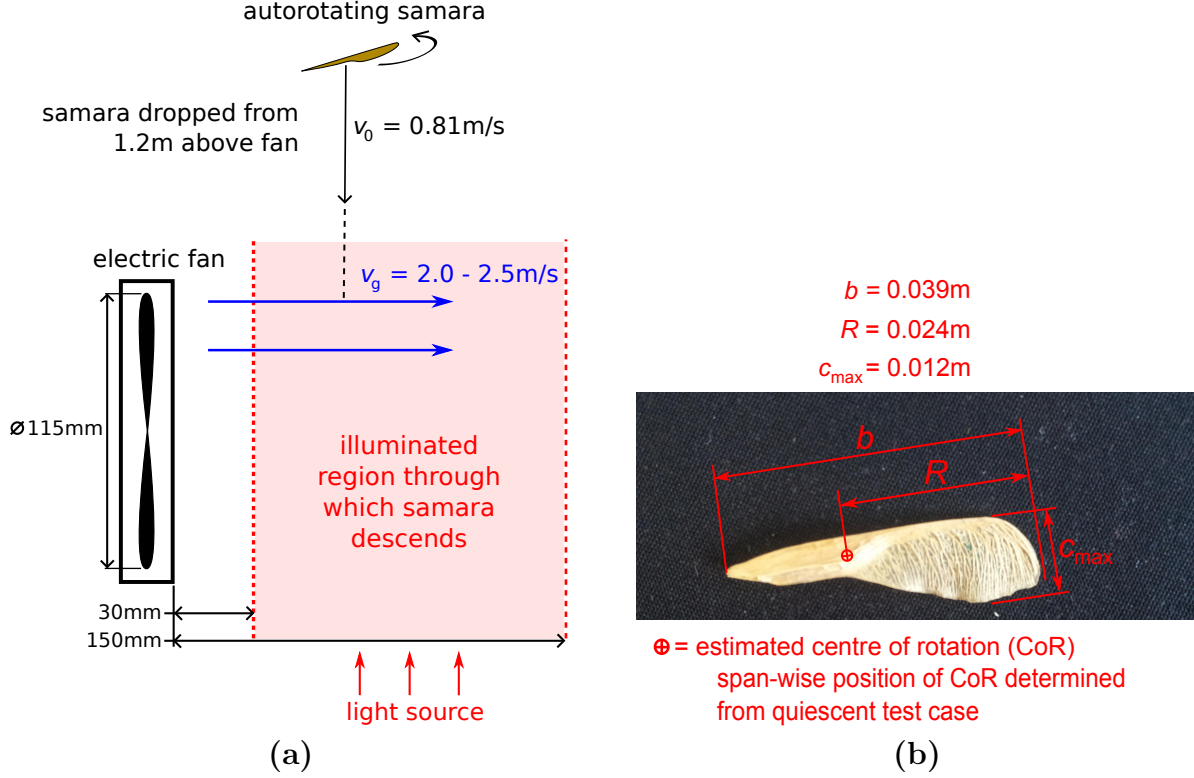


Figure 4.4: (a) Diagram of the experimental setup. The samara was dropped from sufficient height so as to allow the samara to reach equilibrium kinematics prior to hitting the transverse gust. A video of the samara in quiescent air was used to establish the equilibrium kinematics. (b) *Acer negundo* samara used in the videos. The span-wise CoR location was determined from the video taken in quiescent air. Measured values of total span, b , maximum chord length, $c_{\max}$, and span-wise distance from the CoR to the tip, R , are shown.

in front of the fan at a distance between 30mm and 150mm, as shown in figure 4.4a. In this region, the velocity generated by the fan, v_g , was between 2.0m/s and 2.5m/s, which is greater than two times the equilibrium descent velocity of the samara. The diameter of the fan was 115mm, which is roughly three times the total span of the samara specimen. A light source was placed beneath the descent path of the samara, shining upwards to illuminate it. A black backdrop was placed behind the field of view, such that the illuminated seed stood in stark contrast to the background. The specimen used for both the quiescent and gusting test cases is shown in figure 4.4b.

The span-wise centre of mass (CoM) location is determined from the quiescent-air video by assuming the CoM to be aligned with the centre of rotation (CoR). The equilibrium values

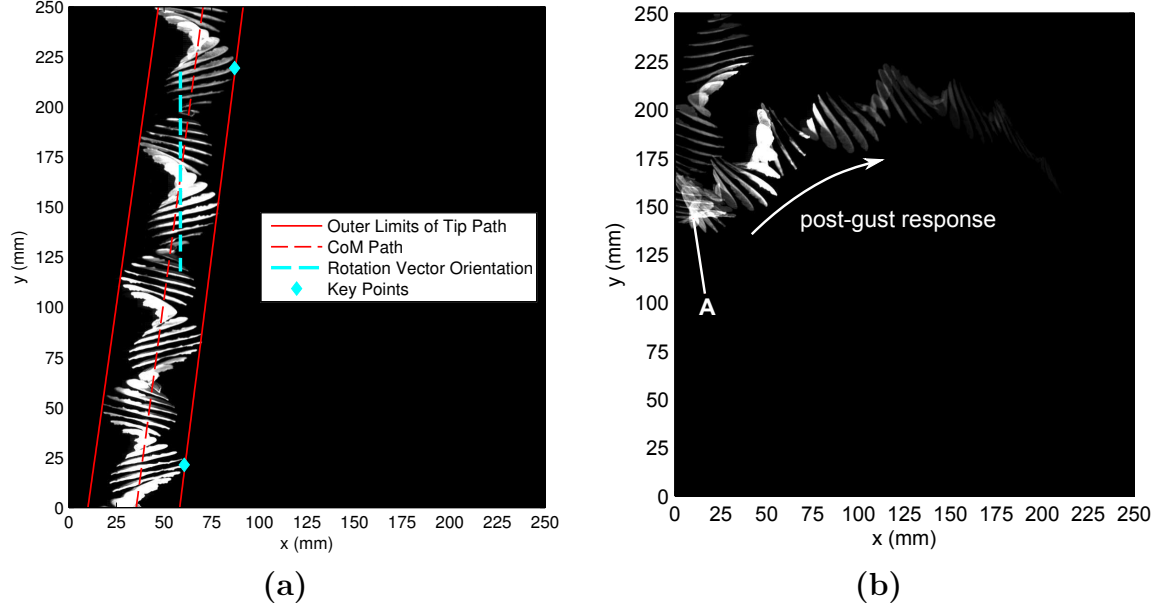


Figure 4.5: Superimposed images from the videos of the samara **(a)** falling in quiescent air and **(b)** falling through a transverse gust from left to right. The equilibrium kinematics of the samara are extracted from the quiescent test case. The inertial response of the samara to the transverse gust is captured by tracking the post-gust displacement of the samara's estimated CoM. The beginning of the post-gust response is taken as the point in time where the CoM is observed to stop momentarily, labelled in the figure as point *A*.

of descent velocity, rotational speed, swept disk radius and coning angle are also extracted from the quiescent test case. In the gusting test case, the post-gust displacement of the CoM is tracked manually by using the estimated span-wise CoM location obtained from the quiescent test case. The time constant that describes its acceleration is extracted by means of exponential regression. This time constant is then compared to the theoretical value given in equation 4.8, which depends on the kinematic parameters of undisturbed autorotation in quiescent air.

4.4 Results

The videos of the falling samara have been provided as supplementary material. By superimposing images that are separated in time by 5ms, the graphics in figures 4.5a and 4.5b were created.

Property	Symbol	Value	Units
Mass	m	50	mg
Span	b	0.039	m
Max chord length	$c_{\max}$	0.012	m
Tip radius from CoR	R	0.024	m
Swept disk radius	R_d	0.023	m
Swept disk area	S	0.0017	m ²
Coning angle	β	14	°
Equilibrium descent velocity	v_0	0.81	m/s
Optimal descent velocity	v_{opt}	0.69	m/s
Rotational speed	Ω	77	rad/s
Tip speed ratio	λ	2.2	

Table 4.2: Physical properties and equilibrium kinematics of *Acer negundo* test samara. Kinematic parameters were extracted from the quiescent test case.

The equilibrium kinematics of the samara are extracted from the video of the samara falling in quiescent air, and are listed in table 4.2. The two key points labelled in blue in figure 4.5a are separated by 492 frames, or 0.246 seconds. Three full rotations of the seed transpire between these key points, yielding an average rotational speed of $\Omega = 77\text{rad/s}$. 797 pixels separate the key points in the vertical direction. The images are calibrated by measuring the total span of the samara, and by determining its length in pixels in the image; this latter measurement is taken when the span-wise axis of the samara is estimated to be normal to the viewing direction. A calibration of $2.49 \times 10^{-4}\text{m/pixel}$ is obtained in this way, yielding an average descent velocity of $v_0 = 0.81\text{m/s}$ between the key points. The optimal velocity, v_{opt} , is found to be 0.69m/s , as calculated using the expression given by Azuma and Yasuda (1989):

$$v_{opt} = \sqrt{\frac{2mg}{\rho S}}. \quad (4.12)$$

As with the species studied by Azuma and Yasuda (1989), the *Acer negundo* specimen approaches optimal autorotation, exhibiting an optimality ratio of $v_{opt}/v_d = 0.86$.

Assuming the CoM to be aligned with the CoR, the CoM trajectory is taken as the central line equidistant from the outermost reaches of the tip path, as shown in figure 4.5a.

This trajectory allows us to estimate the span-wise location of the CoM to be at the 39% span location. The chord-wise CoM position is unknown, but must be nearer to the leading edge than the trailing edge to maintain pitch-stability (Norberg, 1973).

After encountering the transverse gust (from left to right), the samara reorients its axis of rotation and accelerates in a more horizontal direction, as shown in figure 4.5b. The post-gust position of the estimated CoM is tracked manually every 5ms; these location estimates are labelled as *original CoM estimates* in figure 4.6a. The trajectory appears to be parabolic, suggesting that the vertical force component acting on the actuator disk is less than the samara weight in the early gust-response. The original CoM estimates are observed to meander back and forth across the mean parabolic trajectory. This oscillatory deviation can be attributed to error in the estimated CoM location, or to a precession of the rotational axis during the gust-response. These causes cannot be distinguished without an accurate *a priori* knowledge of the CoM location. Instead, the variance of the original estimates about the parabolic curve-fit is taken to represent the uncertainty in the CoM location, δ , as per the relation

$$\delta = 2C \left(\frac{\sum (f_i - y_i)^2}{N} \right)^{1/2}, \quad (4.13)$$

where $f_i - y_i$ is the difference (in pixels) between the original CoM estimates and the parabolic curve in the y-direction. N is the number of points in the original sample, and C is the previously noted calibration constant in metres per pixel. The calculated uncertainty is $\pm 3.3\text{mm}$. Refined estimates of CoM position, made every 2.5ms, are taken to lie on the parabolic arc fitted to the original estimates. Displacement, as measured from the start of the post-gust response, is plotted in figure 4.6b, with the errorbars representing $\pm \delta$.

The theoretical post-gust velocity of the samara, starting from rest, is given as a function of the inertial time constant, τ_i :

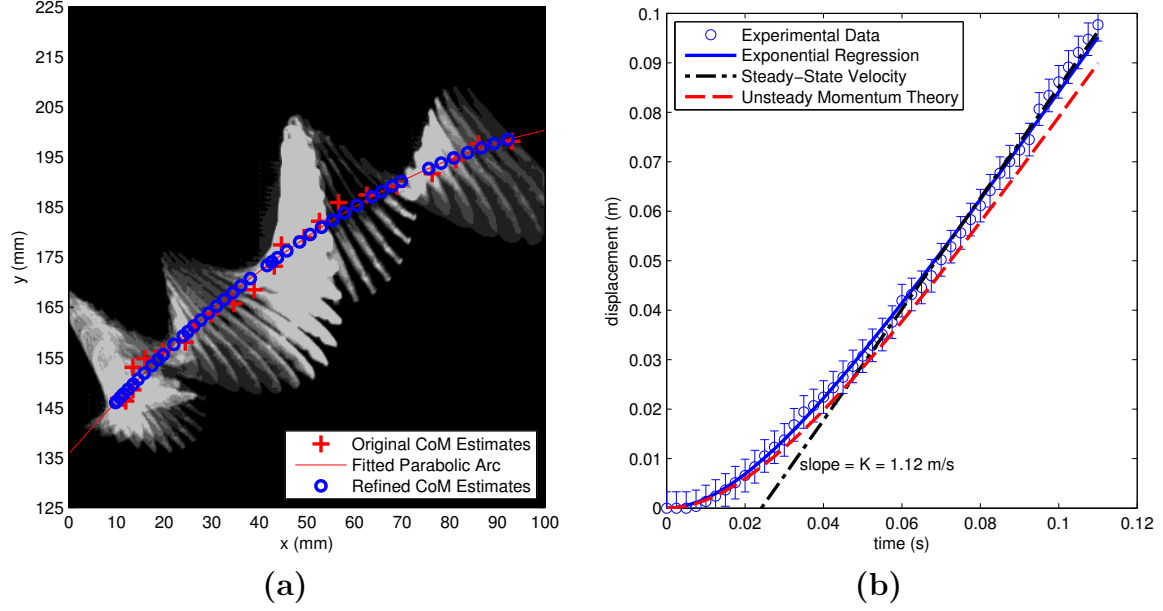


Figure 4.6: **(a)** The post-gust response of the samara is shown with superimposed frames from the video separated in time by 2.5ms. The CoM location was manually tracked every 5ms to produce the original estimates. A parabolic arc was fitted to these estimates, and a set of refined CoM location estimates were taken to lie on this arc. **(b)** The post-gust displacement, as extracted from the figure in (a), plotted versus time. Unsteady momentum theory overpredicts the observed inertial time constant by 20%, as determined by exponential regression of the experimental displacement data. The uncertainty in the displacement, as shown with errorbars, is based on the variance of the original estimates relative to the parabolic curve fit in (a).

$$v(t) = K[1 - \exp(-t/\tau_i)], \quad (4.14)$$

where t is time. The value of K is not known *a priori*, as the gust profile was not characterized. The post-gust displacement eventually approaches a linear function of time, the slope of which is equal to $K = 1.12\text{m/s}$. By using the experimentally-obtained equilibrium velocity in the model, only the theoretical inertial time constant remains to be validated. Integrating equation 4.14 over time, we arrive at an expression for total post-gust displacement along the parabolic trajectory, s :

$$s(t) = K[t + \tau_i(\exp(-t/\tau_i) - 1)]. \quad (4.15)$$

The calculation of the inertial time constant requires the properties and equilibrium kinematics of the test samara, as reported in table 4.2. Taking the density of air to be $\rho = 1.2 \text{ kg/m}^3$, we calculate the theoretical inertial time constant in equation 4.8 to be $\tau_i = 30.6\text{ms}$. The theoretical model now becomes

$$s(t) = 1.12[t + 0.0306(\exp(-t/0.0306) - 1)], \quad (4.16)$$

with displacement and time expressed in metres and seconds, respectively. As shown in figure 4.6b, this theoretical model agrees quite well with the experimental data. An exponential regression analysis was performed on the experimental displacement data, yielding an experimental time constant of 25.5ms. The theoretical time constant of 30.6ms represents a 20% overprediction.

4.5 Discussion

The discussion is separated into two subsections. Firstly, the comparison of momentum theory to the observed gust-response will be discussed in further detail. The theoretical model will be used to consider the gust-characteristics that are most likely to interrupt autorotation. Secondly, the observed gust-response will be discussed in relation to the ecology of samara-producing trees.

4.5.1 Application of momentum theory to the transient dynamics of autorotation

Momentum theory was applied to develop a model for the post-gust displacement of an autorotating samara. The coning angle was assumed to remain constant during the gust-response, as was observed qualitatively in the video (refer to figure 4.5b). The induced velocity was also assumed to be constant, in order to maintain the linearity of the expression for thrust acting on the samara. These simplifications reduce the expression for thrust to that of a steady-state problem. Nonetheless, the model gives a reasonable approximation of

a samara's inertial time constant, overpredicting the experimentally-obtained time constant by 20%. The accuracy of the model is limited by approximating the transverse gust-response as a purely axial gust-response; although the realignment of the rotational axis does generate a sudden change in disk-normal fluid velocity, greater agreement between model and experiment may be observed using an apparatus that generates purely axial gust profiles. At a minimum, however, we can conclude from this experiment that the model is a suitable order-of-magnitude approximation for the inertial time constant. Having engendered some confidence in the model, its implications for the stability of autorotation will be discussed.

In the natural environment, samaras will encounter gusts of varying strength and frequency. The period of such a gust, τ_g , determines whether the problem can be treated as quasi-steady. We will consider a sinusoidal axial gust of the form

$$V_\infty(t) = \begin{cases} \frac{1}{2}V_p \left[1 - \cos \left(2\pi \frac{t}{\tau_g} \right) \right], & t \in [0, \tau_g] \\ 0, & t > \tau_g, \end{cases} \quad (4.17)$$

where V_p is the peak velocity of the gust. Consider the ordinary differential equation that describes the system, as given in equation 4.11 and rewritten here:

$$\tau_i \dot{v}_d + v_d = v_0 - V_\infty,$$

recalling that v_0 is the equilibrium descent velocity of the samara in quiescent conditions. This system is stable, and always tends towards a descent velocity that satisfies

$$v_d + V_\infty = v_0. \quad (4.18)$$

The rotational rate of the samara will similarly tend towards an equilibrium, such that the whole system tends towards a constant tip speed ratio, as defined in equation 4.1 and rewritten here:

$$\lambda = \frac{\Omega R_d}{v_d + V_\infty}.$$

By using the gust profile given in equation 4.17 with a peak velocity of $V_p = v_0$ and applying the initial condition $v_d(0) = v_0$, a solution to the differential equation in 4.11 can be obtained numerically. MATLAB's Runge-Kutta solver, *ode45*, was applied to solve for $v_d(t)$. For discussion purposes, rotational speed will be assumed to be constant, and only the dynamics of the descent velocity will be considered. The deviation of tip speed ratio from its equilibrium value, λ_0 , is then given by

$$\frac{\lambda}{\lambda_0} = \frac{v_0}{v_d(t) + V_\infty(t)}. \quad (4.19)$$

The effect of gust period on tip speed ratio is shown in figure 4.7. For slow gusts, i.e. $\tau_i/\tau_g \ll 1$, the instantaneous value of V_∞ is matched by a decrease in descent velocity that is equal in magnitude. The sum $v_d + V_\infty$ remains nearly constant at all times, and the tip speed ratio is not greatly disturbed from equilibrium, i.e. $\lambda/\lambda_0 \approx 1$. As the gust period decreases, the system can no longer be treated as quasi-steady; a phase lag develops between the freestream velocity and the resultant descent velocity, and deviations from the equilibrium tip speed ratio occur. In response to quick gusts, i.e. $\tau_i/\tau_g \gg 1$, the descent velocity remains roughly constant as the gust passes, and the tip speed ratio varies according to the changes in freestream velocity. The inertial time constant determines the range of gust periods that will disturb the tip speed ratio of the samara from equilibrium. For a given gust period, an increase in peak gust velocity will also increase the deviation from equilibrium tip speed ratio.

When the equilibrium tip speed ratio is disturbed, changes in CoP location will occur; if the chord-wise CoP location varies sufficiently far from alignment with the chord-wise CoM, pitch-stability will be compromised and autorotation will be disrupted. The degree of deviation necessary to interrupt autorotation depends on the moment of inertia about

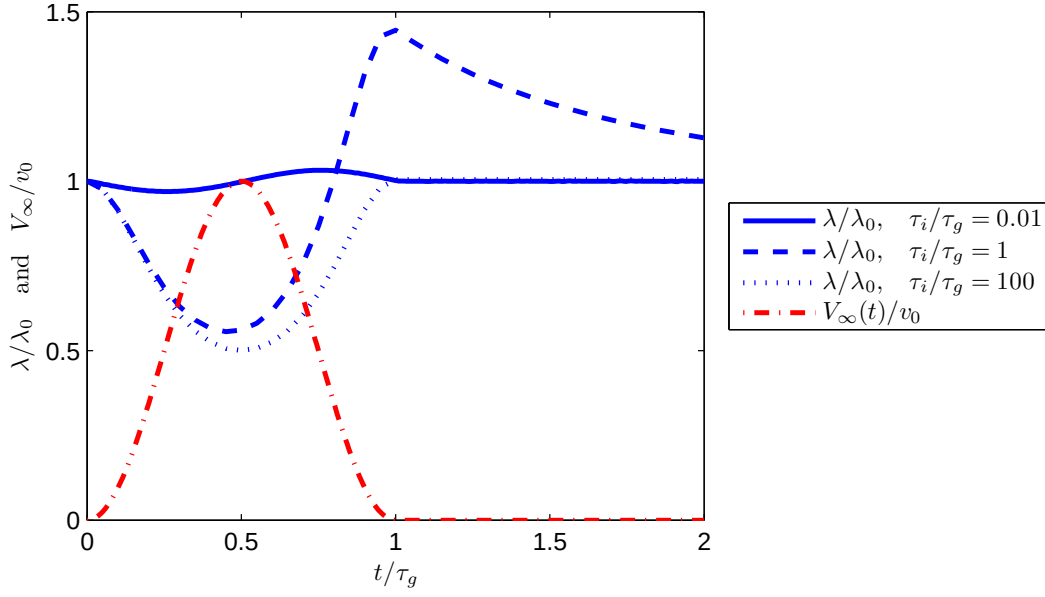


Figure 4.7: The deviation of tip speed ratio from equilibrium, λ/λ_0 , in response to the shown axial gust profile, V_∞/v_0 . When the inertial time constant is much less than the gust period, i.e. $\tau_i/\tau_g \ll 1$, tip speed ratio does not deviate greatly from equilibrium. When the gust period is of similar magnitude or less than the inertial time constant, significant deviations from equilibrium occur, which is liable to disrupt autorotation by compromising pitch-stability.

the pitch axis. To precisely quantify the limits of stable autorotation for a specific samara, further dynamic analysis would be required.

A *sustained* decrease in tip speed ratio could also lead to the detachment of an otherwise stable LEV, as a minimum tip speed ratio is necessary allow LEV attachment in steady-state (see chapter 3). The momentum model could be used to estimate whether the tip speed ratio would drop below a threshold value in response to a given gust. However, a quasi-steady approach to LEV attachment is likely incomplete, and a *transient* decrease to very low tip speed ratios does not necessarily guarantee LEV detachment; the dynamics of stably-attached LEVs remains an open research topic. With regards to autorotation, the detachment of an LEV would remove any stabilizing effect it had on the instantaneous CoP location, possibly also leading to pitch-instability.

4.5.2 Consequences for samara ecology

Given that samaras depend on transverse winds to disperse, their ability to maintain autorotation during sudden changes in horizontal wind velocity is crucial to reproductive success. The *Acer negundo* samara specimen was observed to realign its axis of rotation to be more parallel to the new freestream velocity upon hitting the gust. Such a reaction is made possible by the non-zero coning angle, which presents a blockage to transverse flows to create a flapping moment. This realignment increases the horizontal force experienced by the samara, allowing it to increase its dispersal using the horizontal winds that it encounters. Although beneficial to the reproduction of samara-producing trees, such amplification of horizontal disturbances may or may not be desirable in the design of bio-inspired aerial vehicles.

It is important to recall that the preceding analysis on the stability of autorotation is based on pitch-stability. There are examples of samaras that do not exhibit pitch-stability in their equilibrium flight, such as those of the genus *Fraxinus spp.* Instead, these samaras autorotate about two axes, the vertical and span-wise axes, which is made possible by a CoM that is centred in the chord-wise direction. It has been hypothesized that a benefit of two-axis autorotation is the ability to navigate gusty environments (McCutchen, 1977), unrestricted by the limits of pitch-stability. This hypothesis was directly tested by Greene and Johnson (1990), but they observed no interruption of the autorotation of single-axis samaras in the outdoor environment selected for their study; in gustier conditions, the hypothesis might yet hold. The present work provides a specific physical mechanism with which to compare the stability of these two types of autorotation. By systematically subjecting single- and two-axis samaras to axial gusts of varying frequency and strength, the gust forms which would favour two-axis samaras can be determined. These gust forms can then be compared to the known wind/gust conditions in an ecosystem of interest, in order to predict whether two-axis autorotation would be more robust in that environment.

4.6 Conclusions

Two high-speed videos of an autorotating *Acer negundo* samara were acquired for study. The equilibrium kinematics were extracted from a quiescent test case, where the samara's descent was undisturbed. A transverse gust was then simulated by dropping the samara in front of an electric fan. The samara was shown to exhibit impressively robust autorotation, continuing to rotate during a rapid change in trajectory. The samara realigned its axis of rotation to be parallel to the oncoming freestream flow, thereby increasing its horizontal acceleration and resulting dispersal distance. This behaviour is ecologically beneficial to the reproduction of samara-producing trees, but may be undesirable in biomimetic devices where disturbances may need to be dampened, not amplified.

The transient acceleration of a samara in response to a gust was modelled by treating the samara as an actuator disk and applying momentum theory. By manually tracking the trajectory of the post-gust centre-of-mass location in the high-speed video, the samara was observed to reach a new steady-state velocity after the gust. The time constant associated with its acceleration was overpredicted by the momentum model by 20%.

The momentum model suggests that the inertia of a samara determines the range of gust frequencies that generate a significant deviation from the equilibrium tip speed ratio. Sustained deviations could dislodge a stably-attached leading-edge vortex, which was shown previously to be a feature of samara flight. Changes in tip speed ratio could also compromise pitch-stability by causing a misalignment of the chord-wise centre of pressure and centre of mass.

Acknowledgements

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Chapter 5

Conclusions and outlook

The broad goal of the current work has been to understand the physical mechanisms involved in samara flight. A review of the ecological literature, presented in chapter 2, provided a context from which to form targeted research questions; the aerodynamic studies of chapters 3 and 4 were inspired in this way. The findings of these investigations are summarized here, with reference to biomimetic applications and samara ecology where appropriate. Suggestions for future work are made along the way, as the limits of the current scope are highlighted.

5.1 Leading-edge vortices in samara flight

The kinematic pattern of autorotating samaras bears a visual resemblance to that of a helicopter. Functionally, samaras are most analogous to small helicopters, where flow-separation is commonplace due to the low operating Reynolds numbers. It has been shown previously that the separated flow-field around samaras can exhibit a stably-attached LEV that enhances force production. This additional vertical thrust is beneficial to samara reproduction, as it allows the dispersal of greater seed mass at lower descent velocities. For micro-aerial vehicles (MAVs) with propellers, the stable attachment of LEVs over the blades can improve efficiency; the increased force production would allow for shallower blade angles of attack, improving lift-over-drag ratios to reduce rotor torque at constant thrust production. A comprehensive force model for small propellers undergoing deep stall must account for stable LEV attachment.

In chapter 3, the effect of a freestream flow on LEV formation was studied using particle image velocimetry (PIV) in the water tunnel at the University of Calgary. In samara flight,

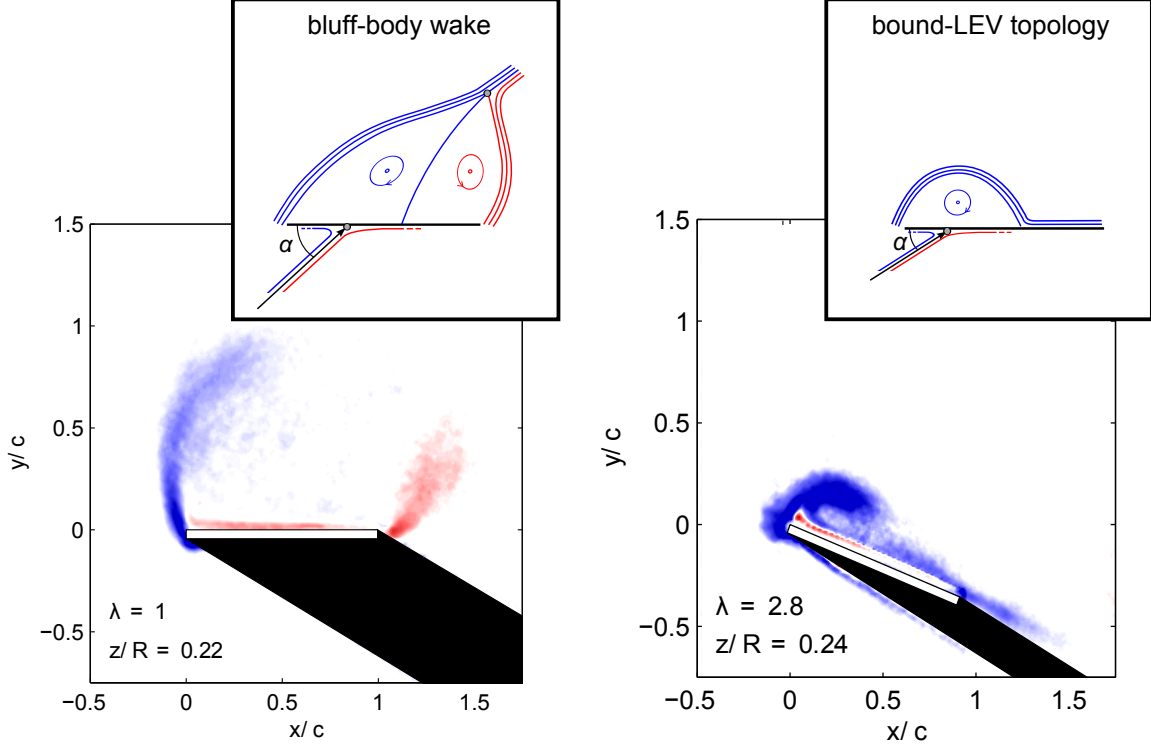


Figure 5.1: Comparison of vorticity distributions and mean flow topology for different tip speed ratios at the most inboard test plane. Increasing tip speed ratio stretches the leading-edge vorticity distribution, making it more compact without changing the integral value of circulation. A minimum tip speed ratio is necessary to generate the bound-LEV topology. The local effective angle of attack, α , is the same for the two test cases shown. Blue and red contours represent opposite-signed vorticity.

this freestream flow represents the descent of the seed. At the Reynolds numbers of samaras - on the order of $O(10^2)$ to $O(10^3)$ - generated forces are very small, presenting a challenge for experimental investigation. Thus, leading-edge circulation and flow topology were studied as proxies for force. Two distinct flow topologies, as viewed on span-normal planes, were identified from the mean velocity fields: that of a bluff-body wake, and that of a bound LEV, as shown in figure 5.1. In the bound-LEV topology, the leading-edge vorticity distribution is concentrated nearer to the plate, increasing the suction peak acting on the plate to enhance force-production. The results of chapter 3 show that a minimum tip speed ratio is necessary to generate the bound-LEV topology. Tip speed ratio is a relevant metric to rotary MAVs undergoing vertical maneuvers, or encountering gusts normal to the rotor disk. The range

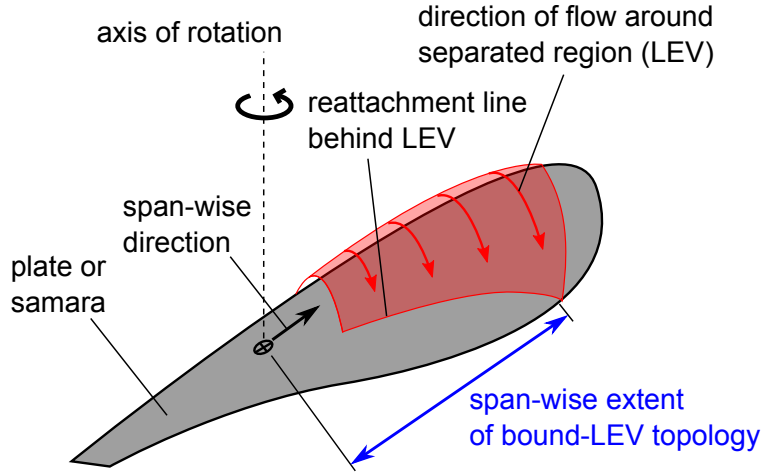


Figure 5.2: Cartoon of the flow-field over a rotating plate exhibiting a stable LEV. Tip speed ratio affects the span-wise extent of the bound-LEV topology; further work is necessary to adequately predict the span-wise extent of the bound-LEV topology under general conditions.

of tip speed ratios exhibited by natural samaras was found to be similar to the range shown to support stable LEV attachment (i.e. the bound-LEV topology), suggesting that many of the considered species do benefit from LEV formation.

Even at sufficiently high tip speed ratio, the bound-LEV topology is only observed at inboard locations (refer back to figure 3.12). The stagnation point behind the LEV is observed to move towards the trailing edge with increasing distance from the root of rotation. At sufficiently outboard locations, the half-saddle is replaced by a full-saddle in the wake, and trailing-edge separation occurs. As the bound-LEV topology is expected to be the most advantageous in terms of force production, further work to predict its span-wise extent under general conditions would be instrumental to the development of a comprehensive force model (see figure 5.2).

Leading-edge circulation on span-normal planes was found to be uniquely defined by the parameters of local effective angle of attack, local effective velocity, and span-wise position; in lifting-line theory, the span-wise circulation distribution is similarly determined for a given planform shape. This surprising analogy is depicted in figure 5.3, but it has some limits; for example, lifting-line theory predicts the total bound circulation around a wing, whereas

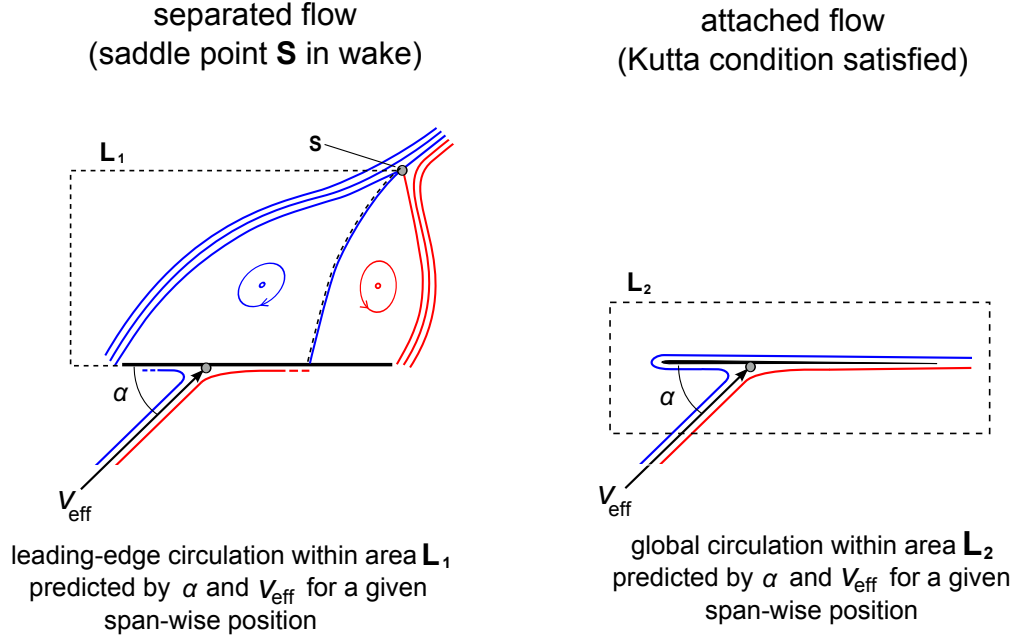


Figure 5.3: At a given span-wise position, local effective angle of attack and local effective velocity determine the global circulation in attached flow (see right); the same inflow parameters determine leading-edge circulation in the separated flows of the current investigation, as calculated from the mean velocity field (see left). Tip speed ratio is important in determining the location of the saddle point.

only the vorticity distribution associated with the leading-edge shear layer and LEV were considered in this work. Another key difference is flow attachment; in traditional lifting-line theory, the flow is generally assumed to be attached, with a fixed saddle point at the trailing edge as per the Kutta condition. For a plate undergoing deep stall, the location of the saddle point in the wake is an additional degree of freedom. Tip speed ratio and span-wise position determine the location of the saddle point on span-normal planes, fixing the topology. Despite the effect of tip speed ratio on flow topology, it has negligible effect on leading-edge circulation. The net effect of increased tip speed ratio on the mean flow-field can be interpreted as vortex stretching, whereby the leading-edge vorticity distribution is made more compact without affecting the integral value of circulation.

The effects of planform design on LEV formation were not investigated in this work, although samaras in nature exhibit a variety of curved planform shapes. It has been shown previously that low aspect ratio is correlated to the observation of stable LEVs (Lentink and

Dickinson, 2009), but the specific profile of the leading and trailing edges could also play a role. For pitching profiles, Hartloper and Rival (2013) showed how leading-edge curvature affects the interaction of the tip and leading-edge vortices; a similar study for steadily-rotating plates would be insightful. The effect of coning angle in natural samara flight is another area for future research. At non-zero coning angles, a portion of the disk-normal velocity is parallel to the span. Such bulk span-wise flow has been shown to mediate LEV growth over rotating plates of high aspect ratio (Wong and Rival, 2013); the steady-state effect of span-wise flow over rotating plates of low aspect ratio remains unknown.

5.2 Gust-response and stability of samara flight

The experimental investigation in chapter 3 was analogous to equilibrium autorotation, with constant values of rotational speed and descent velocity. However, an impressive feature of samara flight is the tendency to maintain autorotation in gusty conditions. In chapter 4, the transient dynamics of autorotors were considered using an *Acer negundo* samara as the subject of investigation. The test specimen exhibited robust autorotation, continuing to rotate through a rapid change in trajectory as it responded to a horizontal gust; this behaviour was observed using high-speed video. Momentum theory was applied to develop a model to describe the gust-response of an autorotor. The model was validated against the test specimen’s actual gust-response, and relatively good agreement was recorded. The derived model is not specific to samaras, but could be applied to a small propeller encountering a gust. The derived model is linear, making it attractive for implementation in control systems. To preserve the linearity of the model, the induced velocity on the swept disk was assumed to remain constant through the gust-response; for autorotors, it is expected that this assumption becomes less valid when the inertial gust-response is slow, and the flow-field relative to the wing experiences a significant disturbance from equilibrium. For more general applications, if the induced velocity cannot be assumed constant, the full non-linear

expression for thrust on an actuator disk should be considered, as developed by Carpenter and Fridovich (1953).

Gusts parallel to the axis of rotation can generate transient changes in the tip speed ratio, which can affect the topology of the flow-field as viewed from a samara-fixed frame of reference. The degree to which gusts perturb the tip speed ratio was shown theoretically to depend on samara inertia. Samaras of sufficiently low inertia respond quickly to changes in the freestream flow, such that the tip speed ratio remains roughly constant and the topology of the flow-field can be expected to remain largely unchanged. For samaras of greater inertia, the same gust may generate larger transients in the tip speed ratio. In a quasi-steady sense, one might expect a large decrease in tip speed ratio to cause an LEV to detach, as it was shown in chapter 3 that a minimum tip speed ratio is necessary to sustain stable LEV attachment in steady-state. However, a quasi-steady description of the problem may be incomplete; the dynamic behaviour of stably-attached LEVs remains an open research topic.

A large disturbance to the instantaneous tip speed ratio can also cause a significant misalignment of the chord-wise CoP and CoM; such misalignment could compromise pitch-stability, interrupting autorotation. The model suggests that increasing the frequency or the peak velocity of a gust will increase the disturbance to the equilibrium tip speed ratio, and therefore increase the likelihood of instability. For a given gust, an increase in samara inertia will generate larger transient changes in the tip speed ratio, but the corresponding effect on the stability of autorotation is unclear; an increase in inertia also affects the dynamic response in pitch, such that larger movements of the CoP may be tolerable for a heavier samara. In order to inform the design of fabricated autorotors, further work is warranted to determine how dynamic stability scales with increasing samara size and weight.

The stability of autorotation is important to the reproduction of samara-producing trees, as maintaining autorotation allows for further seed dispersal. It has been hypothesized previously that samaras with two rotational axes are more stable in gusty conditions (McCutchen,

1977), providing them with a competitive advantage over single-axis samaras. For a specific set of wind conditions, this hypothesis was negated by Greene and Johnson (1990). They did not observe any of the single-axis samaras to cease autorotating during their test, but it cannot be ruled out that gustier conditions might be detrimental to single-axis autorotation. The current work provides a physical mechanism of instability to test this hypothesis further. In chapter 4, it was shown theoretically that pitch-stability could be compromised by axial gusts of sufficient strength or frequency. This mechanism of instability could be systematically tested in a vertical wind tunnel capable of generating sinusoidal gusts of varying frequency and amplitude. As gust frequency or strength increases, it is hypothesized that the autorotation of a single-axis samara becomes more likely to be interrupted. If a combination of gust frequency and peak velocity is identified to disrupt single-axis autorotation, it can be compared to the gust forms in an ecosystem of interest; it can then be determined whether two-axis autorotation would exhibit improved robustness of autorotation in that specific environment. The momentum model presented in chapter 4 would be useful to such a study, providing an estimate of the range of gust frequencies that would generate significant transients in the tip speed ratio.

5.3 Closing remarks

As has been recounted above, the study of samara flight is relevant to the design of small propellers and man-made autorotors. The experimental investigations presented in this thesis represent incremental steps towards the modelling of force-production and gust-response of small rotary aircraft. The complexities of natural samara flight could not all be considered here, and many tangential topics remain for future investigation. The biomimetic approach to engineering design will never want for inspiration, as the vastness of nature has much to offer.

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Appendix A

Sources of error and uncertainty in the PIV investigation

In the experimental investigation of chapter 3, particle image velocimetry (PIV) was used to quantify the velocity field on span-normal planes. The uncertainty in the results were not reported in the main body of the thesis, as well-established estimates of uncertainty for PIV are not readily available. Raffel et al. (2007) considered the RMS random error introduced by individual sources of error, although applying these numerical estimates to general investigations with multiple sources of error is difficult. The key sources of error in the PIV investigation of chapter 3 are discussed here; the various best practices used to minimize these sources of error are also summarized. Based on the work of Raffel et al. (2007), an estimate of uncertainty in the calculated values of circulation is also presented.

A.1 In-plane particle displacement

The frame rate of the camera must be set such that most of the particles in a given interrogation window do not displace out of the window by the time the next image is captured; such loss of particles results in poorer correlations. An appropriate rule-of-thumb is to aim for an interframe pixel displacement of one-quarter of the smallest interrogation window size. The smallest window used in this investigation was 16x16 pixels, and an interframe particle displacement of four pixels was targeted. The typical velocity of a particle was taken to be the effective velocity at the midspan, which was held constant for all three test cases at a value of $v_{\text{mid}} = 0.036\text{m/s}$. The typical particle displacement in pixels, s_p , during the interframe time, Δt , is calculated as

	Frame Rate	s_p
Small FoV	500 Hz	3.4 pixels
Large FoV	125 Hz	3.2 pixels

Table A.1: Frame rate and typical particle displacement (in pixels) for both FoVs. Typical pixel displacement is calculated as per equation A.1.

$$s_p = v_{\text{mid}} \Delta t \frac{1}{C}, \quad (\text{A.1})$$

where C is the calibration constant of the FoV in meters per pixel. The frame rates and corresponding typical displacements for both FoVs are summarized in table A.1.

Obviously the velocity of individual particles will vary; however, by using this quarter-window rule of thumb, good correlations were achieved with clear peaks in the correlation map. The loss of particles to in-plane convection was further reduced by performing multi-pass correlations with decreasing interrogation window size: one pass was completed with windows of 128x128 pixels, and two subsequent passes were completed on windows of 16x16 pixels. In completing multi-pass correlations, the interrogation windows were displaced from one frame to the next according to the calculated mean velocity from the previous pass. This approach reduces the number of particles that are lost to in-plane convection, as the window effectively “follows” the particles.

A.2 Out-of-plane particle losses

In three-dimensional flows, particles can be lost between consecutive images due to out-of-plane displacement. The flow in this investigation was definitely three-dimensional, with expected out-of-plane velocities approaching similar magnitudes as the in-plane velocities in some locations. With a sufficiently thick laser sheet, or a sufficiently quick frame rate, out-of-plane particle losses can be kept low. In this PIV study, the thickness of the laser sheet was of similar magnitude to the width of the smallest interrogation window. For both sampled FoVs, the laser sheet thickness was measured to be 1mm; the width of the smallest

interrogation windows were 0.34mm and 1.5mm for the small and large FoVs, respectively. With the frame rates set sufficiently fast so as to prevent particle losses due to in-plane convection, out-of-plane losses were consequently also kept low.

As with the other sources of error, the uncertainty in the mean velocity fields were kept low by completing an ensemble average of one hundred correlations per test plane. Much of the random error was removed in this way, although the instantaneous variations in the flow-field were also lost. For the purposes of the study, capturing the mean flow-field without recording the local fluctuations was sufficient.

A.3 Particle density

In order to achieve fine spatial resolution, it is advantageous to use the smallest possible interrogation windows. The minimum window size is limited by the density of particles in the fluid. In order to achieve low uncertainty in the velocity vectors, it is best practice to have at least four particle image pairs in the interrogation window in consecutive images (Raffel et al., 2007). To demonstrate that the density of particles in the experiment was sufficient, example images are shown in figure A.1. The sample test case is the twisted plate at a tip speed ratio of $\lambda = 2.8$ at the midspan plane.

In each column of figure A.1, the top image shows a large portion of the FoV, and each subsequent row shows a smaller subset thereof. For the small FoV, the region selected to show a 16x16 pixel interrogation window is in the high-velocity region of interest for quantifying the shear-layer velocity profile. The chosen region is seen to have few bright particles, as shown in the red rectangle in the 128x128 window (middle row, left column). The interrogation window shown in the bottom row (left column) is thus a sort of worst-case scenario; nonetheless, at least four local maxima of brightness indicate individual particles, which is sufficient to achieve a good correlation between consecutive images. It should also be noted that the images shown in figure A.1 have an 8-bit depth for display on a computer

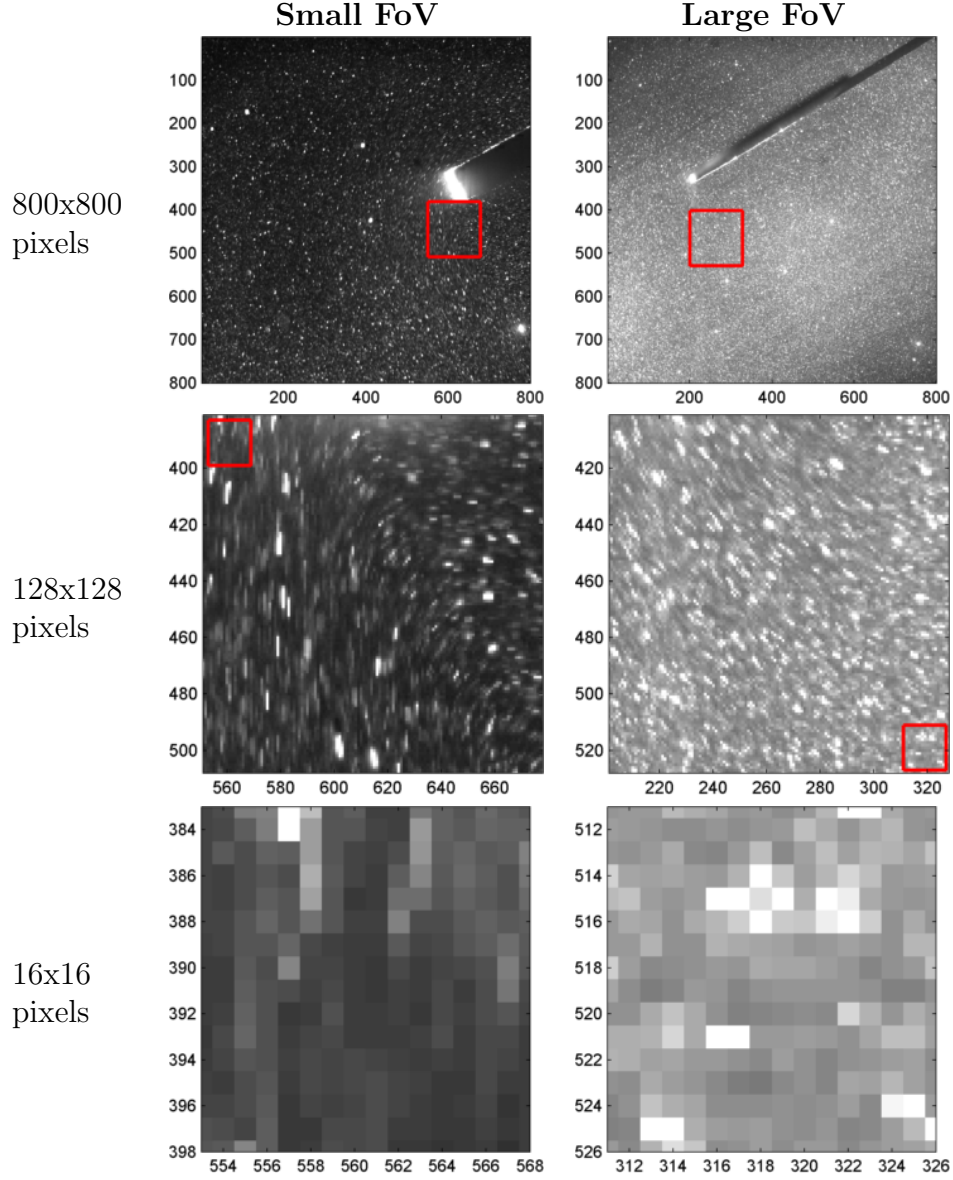


Figure A.1: Sample images from the twisted plate test case at $\lambda = 2.8$ at the midspan location. **Left column:** Small FoV used to quantify shear-layer velocity profile. **Right Column:** Large FoV used to quantify leading-edge circulation. **Top Row:** 800x800 pixel window. Red rectangle highlights the region that is portrayed in the middle row. **Middle Row:** 128x128 interrogation window, as sampled in the first pass of the multi-pass correlation scheme. The red rectangle in these images highlights the region shown in the bottom row. **Bottom Row:** 16x16 interrogation window; this is the smallest interrogation window used. At least four local maxima of brightness are visible for both FoVs, demonstrating sufficient particle density to acquire accurate correlations.

monitor; the actual raw images have a 12-bit depth, such that a greater dynamic range is available to perform correlations than is visible in the figure. This greater depth increases the signal-to-noise ratio.

A.4 Correlation peaks and rejected vectors

The Q-factor is a measure of the magnitude of the detected correlation peak; it represents the ratio between the magnitude of the primary peak to that of the second greatest peak. Higher Q-factor values represent greater signal-to-noise ratios in the calculated correlations. For both the large and small FoV's, the Q-factor was sampled for three arbitrary correlations of the twisted plate test case at the midspan at $\lambda = 2.8$. These values are reported in table A.2. During processing, the software automatically detects spurious vectors and rejects them; they are then replaced by interpolation of the neighbouring data points. The number of rejected vectors are also reported in table A.2. It is clear from table A.2 that the correlation peaks are sufficiently large to avoid ambiguity, and the number of rejected vectors is inconsequential.

	Small FoV	Large FoV
Q-factor	2.60, 2.57, 2.57	2.38, 2.39, 2.40
Rejected vectors	5, 4, 4	0, 0, 0
Total vectors in FoV	65,536	65,536

Table A.2: Q-factor values and number of rejected vectors for spot-checked PIV correlations.

A.5 Estimate of uncertainty in velocity, vorticity and circulation

The sources of error expected to be most dominant are the loss of particles to in- and out-of-plane convection, the variations in particle image diameter, and the presence of displacement gradients due to shear. The resulting total uncertainty in the measured pixel displacements, u_d , is taken to be $u_d = \pm 0.1$ pixels, based on the estimates presented by Raffel et al. (2007). This uncertainty can now be propagated through the calculation of velocity, vorticity and

circulation. Velocity is calculated from the measured pixel displacements, d , as per the relation

$$V = \frac{Cd}{\Delta t}, \quad (\text{A.2})$$

where V is velocity magnitude, C is the calibration constant of the FoV in metres per pixel, and Δt is the interframe time. The uncertainty in this velocity, u_{Vi} , is given by

$$u_{Vi} = \frac{\partial V}{\partial d} u_d = \frac{C}{\Delta t} u_d. \quad (\text{A.3})$$

Vorticity is calculated from velocity gradients. A central difference scheme was performed to achieve numerical differentiation. At any interior point, vorticity is calculated as

$$\omega_i(x, y) = \frac{\partial V_y}{\partial x} - \frac{\partial V_x}{\partial y} \approx \frac{V_y(x + \Delta x) - V_y(x - \Delta x)}{2\Delta x} - \frac{V_x(y + \Delta y) - V_x(y - \Delta y)}{2\Delta y}, \quad (\text{A.4})$$

where $\omega_i(x, y)$ is the vorticity at (x, y) in an individual correlation, and V_x and V_y are the velocities in the x and y directions, respectively. Δx and Δy are the distances between measured velocity vectors in the x and y directions, respectively. When using a central difference scheme, Raffel et al. (2007) presents the uncertainty in a calculated velocity derivative to be

$$u_{\text{derivative}} = \frac{0.7u_{Vi}}{h}, \quad (\text{A.5})$$

where h is the spacing between measured vectors in the direction of the derivative, and $\Delta x = \Delta y = h$. This uncertainty is obtained through standard uncertainty-propagation techniques, as used here in appendices A and B. Vorticity is the difference of two partial derivatives, and the resulting uncertainty in vorticity, $u_{\omega i}$, is

$$u_{\omega i} = \frac{u_{Vi}}{h}. \quad (\text{A.6})$$

The measured velocity vectors are separated by 4 pixels, as 75% overlap was used on the smallest interrogation windows of 16×16 pixels. In metres, the distance between vectors is given as $h = 4C$; combining this expression with equations A.3 and A.6, we obtain

$$u_{\omega i} = \frac{1}{4\Delta t} u_d. \quad (\text{A.7})$$

The preceding analysis estimates the measurement uncertainty for a single correlation; in practice, an ensemble average of multiple test runs, i.e. multiple correlations, is performed to reduce the effect of random errors. By obtaining an average of n test correlations, the uncertainty is reduced by a factor of $\sqrt{n}$, as shown with standard error propagation techniques. The average vorticity at a specific location, $\omega(x, y)$, can be calculated as

$$\omega(x, y) = \frac{1}{n} \sum_{i=1}^n \omega_i(x, y), \quad (\text{A.8})$$

where $\omega_i(x, y)$ is the vorticity at (x, y) from an individual test run. The uncertainty in the vorticity calculated from the ensemble average of all correlations is thus given as

$$u_{\omega} = \frac{1}{n} \left(n u_{\omega i}^2 \right)^{1/2} = \frac{1}{\sqrt{n}} u_{\omega i}, \quad (\text{A.9})$$

and thus

$$u_{\omega} = \frac{1}{4\sqrt{n}\Delta t} u_d. \quad (\text{A.10})$$

It is clear that for a sample size of one, the expression on the right-hand side of equation A.10 simplifies to that in equation A.7. In the investigation of chapter 3, one hundred correlations from individual test runs were averaged for each test plane before plotting the vorticity contours seen in figure 3.8, reducing the local uncertainty by a factor of ten; the uncertainty in mean local velocities, u_V , is likewise reduced by a factor of ten, such that $u_V = 0.1u_{Vi}$. After completing the ensemble average, the signal-to-noise ratio in the calculated velocities and vorticities becomes quite high, as demonstrated for the large FoV test cases in table A.3.

Plate	λ	z/R	V_∞ m/s	u_V m/s	u_V/V_∞ —	$\omega_{\min}^*$ /s	u_ω^* /s	$u_\omega^*/\omega_{\min}^*$ —
Flat	1	0.22	0.031	1.2×10^{-4}	0.004	15.4	0.36	0.023
Flat	1	0.36	0.031	1.2×10^{-4}	0.004	16.8	0.35	0.021
Flat	1	0.50	0.031	1.2×10^{-4}	0.004	15.2	0.33	0.022
Flat	1	0.64	0.031	1.2×10^{-4}	0.004	16.5	0.31	0.019
Flat	1	0.78	0.031	1.2×10^{-4}	0.004	12.4	0.29	0.024
Flat	2.8	0.22	0.021	1.2×10^{-4}	0.006	22.9	0.47	0.020
Flat	2.8	0.33	0.021	1.2×10^{-4}	0.006	20.6	0.40	0.019
Flat	2.8	0.51	0.021	1.2×10^{-4}	0.006	15.6	0.31	0.020
Flat	2.8	0.63	0.021	1.2×10^{-4}	0.006	12.8	0.27	0.021
Flat	2.8	0.79	0.021	1.2×10^{-4}	0.006	13.2	0.23	0.017
Twisted	2.8	0.24	0.021	1.2×10^{-4}	0.006	21.0	0.45	0.022
Twisted	2.8	0.37	0.021	1.2×10^{-4}	0.006	17.9	0.38	0.021
Twisted	2.8	0.50	0.021	1.2×10^{-4}	0.006	16.4	0.32	0.019
Twisted	2.8	0.63	0.021	1.2×10^{-4}	0.006	13.4	0.27	0.020
Twisted	2.8	0.79	0.021	1.2×10^{-4}	0.006	11.6	0.23	0.019

Table A.3: Uncertainties in velocity and normalized vorticity for the large FoV test cases. The worst-case uncertainty in velocity is 0.6% of the freestream velocity, and the worst-case uncertainty in normalized vorticity is 2.4% of the peak negative normalized vorticity on any span-normal plane.

In relative terms, the worst-case uncertainty in velocity is 0.6% of the freestream velocity; the uncertainty in normalized vorticity is 2.4% of the peak negative normalized vorticity on any span-normal plane, $\omega_{\min}^*$. For the small FoV test cases, the key source of error that increases in significance is that of particle density. However, as shown in section A.3, it is expected that the particle density is still sufficiently high in the small FoV to prevent larger uncertainties in the measured particle displacements.

Circulation, Γ , is calculated from an area integral of the discrete vorticity distribution using the expression

$$\Gamma = h^2 \sum_{i=1}^N \omega_i, \quad (\text{A.11})$$

where each ω_i is a separate data point of vorticity within the FoV, and N is the total number of data points used in the sum operation. The uncertainty in circulation, u_Γ , is calculated as

$$u_\Gamma = h^2 \left(N u_\omega^2 \right)^{1/2}. \quad (\text{A.12})$$

Combining equations A.11 and A.12, we obtain an expression for the relative error in circulation:

$$\frac{u_\Gamma}{\Gamma} = \frac{\sqrt{N} u_\omega}{\sum_{i=1}^N \omega_i}. \quad (\text{A.13})$$

Noting that the mean value of vorticity on a given test plane, ω_{mean} , is equal to

$$\omega_{\text{mean}} = \frac{1}{N} \sum_{i=1}^N \omega_i, \quad (\text{A.14})$$

we can express the relative error in circulation as

$$\frac{u_\Gamma}{\Gamma} = \frac{u_\omega}{\omega_{\text{mean}} \sqrt{N}}. \quad (\text{A.15})$$

Since the uncertainty in the vorticity data represents random noise, the integral operation over many vectors causes much of the noise to cancel. As the number of summed data points increases, i.e. as spatial resolution improves, the relative error in circulation decreases. Values of normalized leading-edge circulation and its associated measurement uncertainty are reported in table A.4. Note that the relative uncertainty in dimensional and normalized circulation are identical, such that

$$\frac{u_\Gamma}{\Gamma} = \frac{u_\Gamma^*}{\Gamma^*}, \quad (\text{A.16})$$

where $u_\Gamma^* = u_\Gamma / (v_{\text{eff}} c)$. Normalized circulation was plotted versus local effective angle of attack in figure 3.7; that data is replotted in figure A.2 with vertical error bars denoting the uncertainty as calculated using equations A.10 and A.15. The error bars are smaller than the data markers. The uncertainty in the reported angles of attack is more significant; that uncertainty analysis is presented in appendix B.

Plate	λ	z/R	Γ^*	u_Γ^*	u_Γ^*/Γ^*
Flat	1	0.22	2.04	0.009	0.004
Flat	1	0.36	1.99	0.008	0.004
Flat	1	0.50	1.83	0.008	0.004
Flat	1	0.64	1.57	0.007	0.005
Flat	1	0.78	1.18	0.007	0.006
Flat	2.8	0.22	1.65	0.011	0.007
Flat	2.8	0.33	1.57	0.010	0.006
Flat	2.8	0.51	1.29	0.008	0.006
Flat	2.8	0.63	1.20	0.006	0.005
Flat	2.8	0.79	0.94	0.005	0.006
Twisted	2.8	0.24	1.96	0.012	0.006
Twisted	2.8	0.37	1.95	0.010	0.005
Twisted	2.8	0.50	1.84	0.008	0.004
Twisted	2.8	0.63	1.46	0.007	0.005
Twisted	2.8	0.79	1.11	0.006	0.005

Table A.4: Measured values of normalized leading-edge circulation, Γ^* , and associated uncertainty, u_Γ^* . The worst-case relative uncertainty is 0.7% of the calculated value of Γ^* .

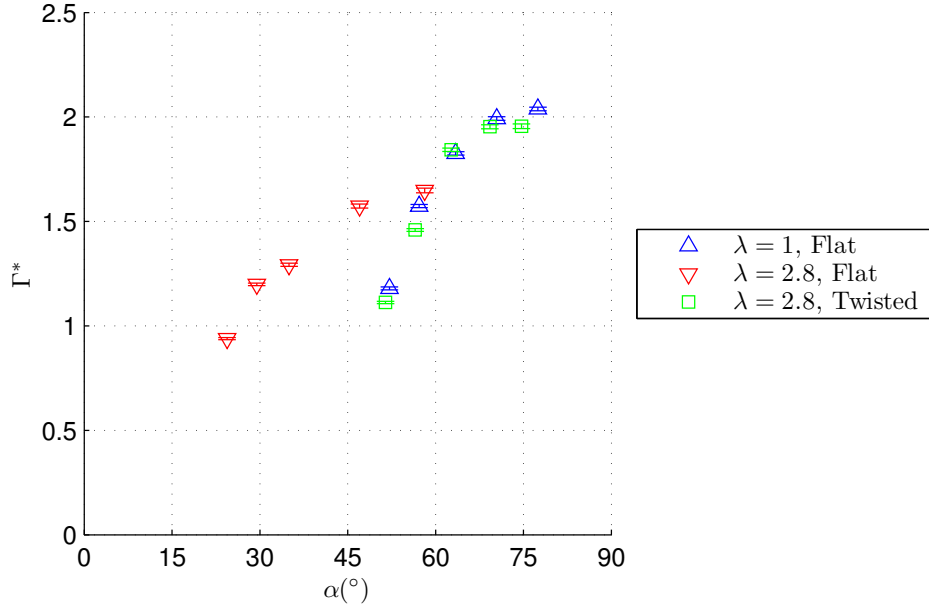


Figure A.2: Normalized circulation, Γ^* , versus local effective angle of attack, α . The vertical error bars denote the estimated uncertainty in the normalized circulation, as calculated using equations A.10 and A.15. The error bars are smaller than the data markers, which makes them difficult to distinguish.

In some applications, the standard deviation of measured values could be used as a measure of uncertainty. Indeed, the sources of error noted above would contribute to increased variance about the calculated mean leading-edge circulation. However, the real instantaneous value of circulation can vary about the mean value, such that the observed variance could be greater than that due to stochastic noise alone. The presented approach to estimating uncertainty avoids this bias by starting from predicted errors in pixel displacement calculations. As a comparison, the worst-case uncertainty obtained using equation A.15 is 0.7% of the calculated value of circulation, whereas the standard deviations of circulation for each test plane lie within 5% and 13% of the calculated value. This scatter could represent real variation in the value of circulation, or could be due to other sources of error not considered herein; without time-resolved data, these effects cannot be distinguished definitively.

Appendix B

Estimation of the uncertainty in the reported angles of attack

In chapter 3, normalized circulation was plotted versus local effective angle of attack in figure 3.7. It was observed that tip speed ratio has negligible effect on normalized circulation at a constant span-wise position and local angle of attack. In order to validate this conclusion further, the uncertainty in the abscissa values - the angle of attack - will be quantified. Local effective angle of attack is calculated as per equation 3.2, rewritten here:

$$\alpha = \arctan\left(\frac{V_{\infty}}{\Omega z}\right) + \tau(z).$$

The uncertainty in the calculated angle of attack is dependent on the uncertainty in the rotational speed of the plate, the freestream velocity, the span-wise position, and the span-wise twist distribution.

The rotation of the plate was driven by a small DC motor with proportional-integral (PI) feedback control. Position feedback was provided by a magnetic encoder, supplying a fine resolution of 2736 counts per revolution. The uncertainty in the measured motor speed was determined by running the motor for two minutes while it remained connected to the flat plate, causing it to rotate in quiescent water. The rotational speed was measured by extracting the frequency of the pulse signal produced by the encoder; this was achieved through a built-in LabView function that performs Fourier analysis. The signal was sampled at 1kHz, and average frequency values were recorded over 0.1s intervals; ten data points per second were collected in this way. The uncertainty is taken as twice the standard deviation of the recorded speed during the two-minute test. This test was repeated for two rotational speeds, 0.419rad/s and 0.661rad/s, yielding an uncertainty of ± 0.04 rad/s in both cases. Bias

z (mm)	0	5	10	15	20	25	30	35	40	45	50	70	90
u_τ (°)	1.2	1.2	3.0	3.9	1.8	2.6	1.6	1.1	1.2	1.2	1.2	1.2	1.2

Table B.1: Uncertainty in measured twist angle as a function of span-wise position

errors in the mean rotational speed were measured to be an order of magnitude lower than the stated uncertainty, and thus have been neglected.

The water tunnel was operated near the low end of its capable range of velocities. The precision of the velocity measurement is 0.001m/s, which amounts to a zero-order uncertainty of ± 0.0005 m/s. In terms of relative error, this is worst for the slowest test case of $V_\infty = 0.021$ m/s, for which the precision of the instrument introduces a 2.4% error.

The span-wise position for each test case was calculated as the difference between the measured heights of the shaft centre-line and the horizontal laser sheet (see figure 3.3b); these heights were measured relative to the bottom of the water tunnel. The zero-order uncertainty of each of these measurements is ± 0.5 mm. When the difference between the measurements is taken, the worst-case uncertainty in the calculated result is ± 1 mm.

The span-wise twist distribution was measured using a digital inclinometer application on a smartphone. The precision associated with this measurement technique is 0.1° , but this does not account for any uncertainty associated with aligning the device at the appropriate span-wise position, and aligning it flush with the plate. The measurement of twist angle was taken five times per span-wise position, yielding a spread of values. The uncertainty based on these measurements, u_τ , was calculated as

$$u_\tau = \pm 2t_4 s_\tau, \quad (\text{B.1})$$

where t_4 is the t-value from the student t-distribution for four degrees of freedom, and $t_4 = 2.776$. s_τ is the calculated standard deviation of the five-measurement sample. The uncertainty in twist angle at various span-wise positions is thus given in table B.1.

The uncertainty in all of the direct measurements can be propagated through the calcu-

lation of angle of attack as per equation 3.2. The resulting uncertainty in angle of attack, u_α , is given by

$$u_\alpha = \left[(\theta_\Omega u_\Omega)^2 + (\theta_v u_v)^2 + (\theta_z u_z)^2 + (\theta_\tau u_\tau)^2 \right]^{1/2}, \quad (\text{B.2})$$

where u_Ω , u_v , and u_z are the uncertainties in rotational speed, freestream velocity, and span-wise position, respectively. θ_i denotes the sensitivity indices, with the subscripts representing the corresponding measured quantities. The indices are calculated as

$$\theta_i = \frac{\partial \alpha}{\partial x_i}, \quad (\text{B.3})$$

where x_i denotes the measured quantity of interest. The resulting expressions for the sensitivity indices are

$$\theta_\Omega = \frac{-V_\infty/z}{(V_\infty/z)^2 + \Omega^2}, \quad (\text{B.4})$$

$$\theta_v = \frac{\Omega z}{(\Omega z)^2 + V_\infty^2}, \quad (\text{B.5})$$

$$\theta_z = \frac{-V_\infty/\Omega}{(V_\infty/\Omega)^2 + z^2}, \quad (\text{B.6})$$

$$\theta_\tau = 1. \quad (\text{B.7})$$

The uncertainty in angle of attack is calculated using equation B.2 for each sampled test plane. The uncertainty in twist angle at each span-wise position is interpolated from table B.1.

Figure 3.7 is replotted in figure B.1 with horizontal error bars to indicate the calculated uncertainty in the reported effective angle of attack. This uncertainty could explain some of the scatter in the data, but is not so large as to call the observed trends into question. The observed collapse of the data (for the flat and twisted plates at tip speed ratios of 1 and 2.8, respectively) does not appear to be coincidental.

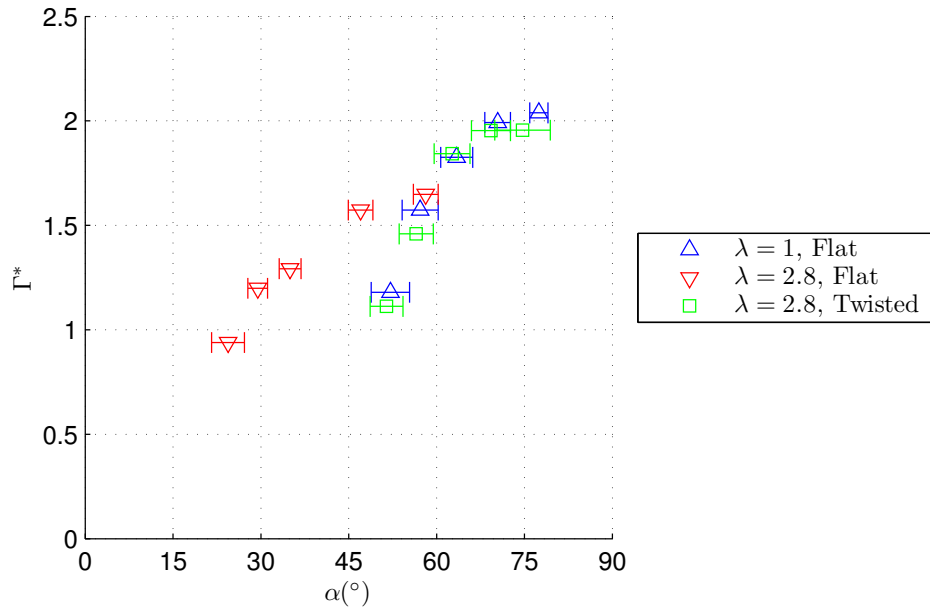


Figure B.1: Normalized circulation, Γ^* , versus local effective angle of attack, α . Uncertainty in the reported angle of attack is represented by the horizontal error bars.

Appendix C

MATLAB code for the calculation of leading-edge circulation

The MATLAB code used to calculate leading-edge circulation on each large FoV test plane is included here. *pivtomat.m* imports the PIV correlation data as exported in ASCII format (in .txt files) by LaVision DaVis 8.1.3. *avgcorr.m* calculates average velocity fields from a set of correlations. *circareaintegral.m* numerically calculates the vorticity distribution of the mean velocity field, then calculates the leading-edge circulation through numerical integration.

pivtomat.m

```
%Import PIV correlations, and write them to individual mat files
%'parent_dir' stores folders of data (in .txt form), 'save_dir' will hold
%folders of mat files. 'identifier' is a string that determines which
%folders will be sampled for ASCII data in .txt format.
%Assumes 256x256 field of data.
%Data format in txt file is as exported by LaVision Davis 8.1.3

function pivtomat(parent_dir, identifier, save_dir)

currdir = cd;

cd(parent_dir);
folder_list = dir(identifier);

for j = 1:length(folder_list)
    cd(folder_list(j).name);
    file_list = dir('*.txt');
    cd(save_dir);
    mkdir(folder_list(j).name);
    cd(parent_dir);

    for k = 1:length(file_list)
        disp(sprintf('%s%s%s', folder_list(j).name, ' FILE #', num2str(k)));
        cd(folder_list(j).name);
        clear X Y U V;
        X = []; Y = []; U = []; V = [];
        [X, Y, U, V] = subimport(file_list(k).name);
```

```

        cd(save_dir);
        cd(folder_list(j).name);
        save(sprintf('%s%s', 'X', num2str(k)), 'X');
        save(sprintf('%s%s', 'Y', num2str(k)), 'Y');
        save(sprintf('%s%s', 'U', num2str(k)), 'U');
        save(sprintf('%s%s', 'V', num2str(k)), 'V');
        cd(parent_dir);
    end
end

cd(currdir);

end %pivtomat

%%
function [X2,Y2,U2,V2] = subimport(filename)

temp = importdata(filename, '\t',1);

s = length(temp.data(:,1));
sqdim = sqrt(s);

clear X2 Y2 U2 V2;
X2 = []; Y2 = []; U2 = []; V2 = [];
if s == 65536 %Assumes 256x256 field
    for i = 1:length(temp)
        X2 = reshape(temp.data(:,1),sqdim,sqdim);
        Y2 = reshape(temp.data(:,2),sqdim,sqdim);
        U2 = reshape(temp.data(:,3),sqdim,sqdim);
        V2 = reshape(temp.data(:,4),sqdim,sqdim);
    end
end

end %subimport

```

avgcorr.m

```

% avg_corr(parent_dir, save_dir, identifier, N)
% parent_dir is where the mat files are stored.
% save_dir is where to save the averaged correlations.
% identifier is a string that determines folders to include.
% N is the number of correlations to skip in each folder for averaging.

function counter = avgcorr(parent_dir, save_dir, identifier, N)

directories;

currdir = cd;
cd(parent_dir);
folder_list = dir(identifier);

```

```

%Hard-coded dimensions of matrices
L = 256;

counter = [];

for i = 1:length(folder_list)
    counter(i) = 0;
    cd(parent_dir);
    cd(folder_list(i).name);

    clear Ua Va;

    X = zeros(L); %initialize X matrix for averaged correlations
    Y = X; U = X; V = X; %initialize other matrices

    for ii = 1:N:(1+99*N)
        disp(sprintf('%s%s%s', folder_list(i).name, ' ', num2str((ii-1)/N+1)))
        clear Xj Yj Uj Vj;
        [Xj Yj Uj Vj] = loadcorrect(ii);

        %Continue if matrices are not missing data
        if (length(Xj)==L) && length(Yj)==L && length(Uj)==L && length(Vj)==L)
            U = U + Uj;
            V = V + Vj;
            counter(i) = counter(i) + 1;
        end
    end

    end

    Ua = U/counter(i);
    Va = V/counter(i);
    Xa = Xj;
    Ya = Yj;

    %Save averaged correlations
    cd(save_dir);
    mkdir(folder_list(i).name);
    cd(folder_list(i).name);
    save('Ua', 'Ua');
    save('Va', 'Va');
    save('Xa', 'Xa');
    save('Ya', 'Ya');
end

cd(homedir);

end

function [X Y U V] = loadcorrect(n)
%Loads mat file and corrects it for plotting with suction side of plate
%facing upwards, and making X & Y proper meshgrid format
%Must already be in directory where .mat file is stored.
clear X Y U V;

```

```

strU = sprintf('%s%s', 'U', num2str(n));
strV = sprintf('%s%s', 'V', num2str(n));
strX = sprintf('%s%s', 'X', num2str(n));
strY = sprintf('%s%s', 'Y', num2str(n));

U = load(strU); U = U.U; %Extract matrix from structure format
V = load(strV); V = V.V;
X = load(strX); X = X.X;
Y = load(strY); Y = Y.Y;

U = U';
V = V';
X = X';
Y = Y';

U = flipud(U);
V = -flipud(V);

end %loadcorrect

```

circareaintegral.m

```

%Calculate circulation using area integral of vorticity
%X,Y are grid matrices (length in mm).
%U,V are velocity matrices (in m/s).
%n is the test case number, used to load the appropriate
%plate location data

function circ = circareaintegral(X,Y,U,V,n)

    c = 36; %chord length in mm

    %Exclude data that is not above the suction side of the plate,
    %or ahead of the leading edge
    directories;
    cd(largeprocessed);
    load('shadowloc');
    W = curl(X/1000,Y/1000,U,V);
    W2 = W .* ( (Y > tan(th1(n)) * X) | (X < 1/tan(th2(n)) * Y) );

    %Ensure erroneous data from camera stamp is removed
    W2 = W2 .* (X < 1.5*c & Y < 1.5*c);

    %Eliminate edges from integration
    W2 = W2 .* (X > -0.5*c & X < 2*c & Y < 1.75*c & Y > -0.5*c);

    %Include only negative-signed vorticity (leading-edge vort)
    W2 = W2 .* (W2 < 0);

    %Integrate numerically
    dx = abs(X(2,2) - X(1,1)); %x increment in mm

```

```
dy = abs(Y(2,2) - Y(1,1)); %y increment in mm
circ = dx/1000 * dy/1000 * sum(sum(W2));

end
```

Appendix D

MATLAB code for extracting rotational frequency of an autorotating samara from high-speed video

In chapter 4, the kinematics of an autorotating samara were extracted from high-speed video. Average values of descent speed and rotational speed were manually extracted as described in section 4.3. For large data sets, this process could be automated. The MATLAB code included here extracts the rotational frequency of a samara as it falls in quiescent air. The samara must be the only illuminated object in the field of view. The code then measures the width of the bright spot, i.e. the projected width of the samara, in every frame, creating a width signal that varies in time. The frequency of rotation is then extracted by means of MATLAB's Fourier transform function, *fft*. Extraction of the descent velocity could be achieved by tracking the centroid of the bright spot in each frame, but that functionality was not completed in the code presented here.

AVIPROCESS.m runs all of the required scripts in the appropriate order. *aviload.m* loads the AVI file into memory. *bgs subtraction.m* removes a known source of erroneous brightness at a consistent location in the images. *avibinary.m* converts the video into a binary video. *seedwidth.m* calculates the width of the projected seed image in every frame; as the samara rotates, the calculated width reaches a maximum and a minimum twice per revolution. *specdens.m* performs Fourier analysis on the seed-width signal and produces a spectral density plot to elucidate the dominant frequency, which is twice the rotational frequency of the samara.

AVIPROCESS.m

```
%Run all scripts necessary to extract kinematic data from video
```

```

aviload; %load avi for processing
bgsubtraction; %replace known bright spots in background with zeros
avibinary; %replace grayscale video with binary video
            %i.e. only two possible values for each pixel
seedwidth; %measure width of seed image
specdens; %perform Fourier analysis on the width signal, and generate the
            %spectral density plot

```

aviload.m

```

% Reads AVI file for processing

% Prompt use to enter full filename in current directory,
% including extension, with single quotes around it.
filename = input('Enter AVI filename');

readerobj = VideoReader(filename);
vidFrames = read(readerobj);
numFrames = get(readerobj, 'NumberOfFrames');

for k = 1 : numFrames
    mov(k).cdata = vidFrames(:,:,k);
    mov(k).colormap = [];
end

disp('Movie imported to variable "mov" for playing with movie command.')
disp('Matrix form of video in variable "vidFrames"')

%Remove duplicate values in vidFrames
vidFrames = vidFrames(:,:,1,:);

```

bgsubtraction.m

```

%Replace a block of an image (or series of images) with zeros (black space)

%Run aviload.m prior to running this script

clear nrows;
% Prompt use to enter number of rows at the bottom of each image
% to convert to black. This was specific to my test setup
nrows = input('Enter number of rows at bottom to convert to black');
vidsize = size(vidFrames);
nframes = vidsize(1,4);

blackrows = zeros(nrows,vidsize(1,2),vidsize(1,3),1);

for i = 1:nframes

```



```

        vidFrames(vidsize(1,1)-(nrows-1):vidsize(1,1),:,:,i) = blackrows(:,:,:,1);
end

```

avibinary.m

```

%converts an image from a frame in an AVI video to a binary image based
%   on input threshold
%plots original image, binary image, bright points and 2D convex hull of
% all bright points

binaryimage = vidFrames;

a = size(vidFrames(:,:,1,:));
nrows = a(1,1);
ncolumns = a(1,2);
nframes = a(1,4);

n = 5; %value to give to non-zero elements in binary image
for f = 1 : nframes
    for i = 1 : nrows
        for j = 1 : ncolumns
            if vidFrames(i,j,1,f) > threshold
                binaryimage(i,j,1,f) = n;
            else
                binaryimage(i,j,1,f) = 0;
            end
        end
    end
end
end
end

```

seedwidth.m

```

%Track width of bright blob in binary video (binary.m must be run first)
clear i f m n;

bysize = size(binaryimage);
nrows = bisize(1,1);
ncolumns = bisize(1,2);
nframes = bisize(1,4);

width = zeros(nframes,1);
left = zeros(nrows,nframes);
right = zeros(nrows,nframes);

%Detect leftmost bright pixel for every row
disp('Starting edge detection')
for f = 1:nframes
    for m = 1:nrows

```

```

        test = 0;
        for n = 1:ncolumns
            if binaryimage(m,n,1,f) ~= 0 && test == 0
                test = 1;
                left(m,f) = n;
            end
        end
    end
end

disp('Left boundary detected')
%Detect rightmost bright pixel for every row
for f = 1:nframes
    for m = 1:nrows
        test = 0;
        for n = 1:ncolumns
            if binaryimage(m,ncolumns - (n-1),1,f) ~= 0 && test == 0
                test = 1;
                right(m,f) = ncolumns - (n-1);
            end
        end
    end
end

disp('Right boundary detected')

%Replace zero values in "left" with large values to find true minimum of
%bright image boundaries
sizel = size(left);

for f = 1:nframes
    for i = 1:sizel
        if left(i,f) == 0
            left(i,f) = sizel(1,1);
        end
    end
end

disp('Calculating width for every frame')
%Determine max width of bright part of image for each frame
counter = 1;
counter2 = 1;
for f = 1:nframes
    width(f,1) = max(right(:,f)) - min(left(:,f));
    if width(f,1) == -sizel(1,1)
        width(f,1) = 0;
    end
    if counter == 100
        msg = [num2str(counter2) '00 frames done'];
        disp(msg)
        counter = 0;
        counter2 = counter2 + 1;
    end
    counter = counter + 1;
end

```

end

specdens.m

```
%Plot the spectral density of the frequencies within a function
clear f fmin fmax dt df Y N NFFT Pyy;

sig = input('Type in name of the vector containing the time signal ');
dt = input('What is the time step between values in the signal? ');
N = max(size(sig))-1;
samplefreq = 1/dt;
df = 1/((N-1)*dt);

NFFT = 2^nextpow2(N);
Y = 2/N * fft(sig,NFFT);

%Calculate power of various frequencies
Pyy = Y.*conj(Y);
fmin = -samplefreq/2;
fmax = -fmin;
f = linspace(0,fmax,NFFT/2+1);
plot(f,Pyy(1:NFFT/2+1),'b')
title('Power Spectral Density')
xlabel('Frequency (Hz)')
```